Educational Investment in Spatial Equilibrium: Evidence from Indonesia

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Governments build schools in particular places, but human capital is portable. This paper asks who benefits from school construction and where the gains accrue. I estimate a spatial equilibrium model of schooling and migration with quasi-experimental variation from Indonesia's *Sekolah Dasar Inpres* program, one of the largest school construction programs in history. First, I document large spatial spillovers. When a district builds a new school, 87.2% of the resulting wage gains accrue outside its borders. Second, gains for people and places diverge. The program raised aggregate output by 2.58% and reduced inequality across people by 1.20%, but also increased inequality across places by 1.26%. Third, alternative allocations can improve equity. Shifting *Inpres* schools toward low-enrollment districts would have reduced inequality across both people and places, while lowering aggregate output only modestly.

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1 Introduction

Governments build schools in particular places, but human capital is portable. Schools serve students locally, and students take their skills wherever they choose to work. When investing in education, governments must consider how schooling and migration decisions jointly shape the distribution of gains across people and places. This paper asks who benefits from school construction, where the gains accrue, and what this implies for where schools should be built.

I answer these questions with national socioeconomic data, quasi-experimental variation from a large-scale school construction program, and a spatial equilibrium model that isolates mechanisms. My goal is to quantify how migration shapes the aggregate and distributional impacts of school construction. My empirical setting is Indonesia, where the *Sekolah Dasar Inpres* program aimed to build 62,000 new primary schools from 1973 to 1978. The program targeted low-enrollment districts with the stated goals of rural development and regional equity. I measure schooling, wage, and migration outcomes four decades later with data from the 2011 to 2014 waves of the National Socioeconomic Survey (*Susenas*).

I evaluate the long-run effects of the program with a difference-in-differences approach. As in Duflo (2001), I compare exposed and unexposed cohorts born in districts with high and low levels of school construction. Exposed cohorts were young enough to attend the new primary schools, while unexposed cohorts were too old. I find that each additional *Inpres* school per 1,000 children raises schooling by 0.148 years, wages by 0.031 log points, and migration from the district of birth by 1.23 percentage points. School construction affects both the human capital that individuals acquire and where they subsequently work.

I capture these responses with a spatial equilibrium model of schooling and migration. Individuals invest in schooling and then migrate for employment. School construction reduces schooling costs and encourages schooling, which increases productivity and wages. Schooling also lowers migration frictions, allowing individuals to reach labor markets where their schooling is better rewarded. In turn, these opportunities encourage further investment in schooling. That is, schooling and migration are complements: schooling facilitates migration, and migration opportunities encourage

schooling. As changes in schooling and migration shift the supply of human capital across locations, wage rates adjust to clear local labor markets. Migration links these labor markets in equilibrium, and so school construction in one district can affect wages in other districts.

I estimate the model using *Inpres* program variation to discipline individual responses and equilibrium adjustments. I reproduce the causal estimates of schooling, wage, and migration effects by applying the same difference-in-differences regressions to model-implied responses. I assess equilibrium adjustment by studying the wage responses of unexposed cohorts as exposed cohorts enter the labor market over time, as in Duflo (2004). The estimates are imprecise but point to moderate congestion. I then use the estimated model to quantify the aggregate and distributional impacts of school construction, tracing gains across people and places. The model allows me to separate underlying mechanisms and evaluate alternative allocations. The regressions instead estimate the combined effects of the program as implemented.

Counterfactuals yield three main findings. First, school construction generates large spatial spillovers. Starting from observed construction, I simulate the addition of one school in each district separately. Summing across these simulations, I find that 87.2% of net wage gains are earned outside the districts that receive additional schools. At the same time, the benefits primarily accrue to the people born in these districts. People born elsewhere instead experience losses, as increased human capital raises the supply of productive labor and reduces equilibrium wage rates. Local wage gains thus capture only part of the effects of school construction.

Second, gains for people and places diverge. Relative to no construction, the *Inpres* program raised aggregate output by 2.58% and reduced inequality across people by 1.20%, but increased inequality across places by 1.26%. Inequality across people compares mean wages by district of birth, while inequality across places compares mean wages by district of work. I use the model to decompose these effects. Migration raises aggregate output and reduces inequality across people, while increasing inequality across places. The reason is that migration helps equalize access to earning opportunities, but it also concentrates human capital in high-wage destinations. Furthermore, the feedback from migration opportunities to schooling amplifies output gains, as better employment prospects encourage further investment in schooling. Equilibrium adjustments dampen these gains and partly reverse the distributional

effects of migration, as increased labor supply lowers wage rates in the places that attract human capital.

Third, alternative allocations can improve equity. I hold the national construction budget fixed and allow up to 20% of observed spending to be reallocated across districts. Allocations that minimize inequality across either people or places reduce both measures, with only small reductions in aggregate output. These results support the program's original targeting of low-enrollment districts on equity grounds. By contrast, output maximization shifts construction away from low-enrollment districts and increases inequality across both people and places. Thus, the historical divergence between people and places does not preclude equity improvements on both margins. The government can pursue these equity goals while preserving most of the aggregate gains from school construction.

My main contribution is to quantify how migration shapes the spatial incidence of educational investment. I bring together a rich set of empirically relevant mechanisms in a spatial equilibrium model of schooling and migration: wage returns to schooling (Mincer 1958), gains from migration (Beegle et al. 2011, Bryan et al. 2014), schooling that increases mobility (Malamud and Wozniak 2012, Machin et al. 2012, Amior 2024), migration opportunities that encourage schooling (Adukia et al. 2020, Khanna and Morales 2024, Ghose 2026), and general equilibrium responses (Khanna 2023, Fujimoto et al. 2025, Chyn and Daruich 2025, Gethin 2025). These mechanisms jointly determine the aggregate gains from school construction and the distribution of its benefits across people and places.

I study school construction as a place-based policy (Glaeser and Gottlieb 2008, Kline and Moretti 2014, Neumark and Simpson 2015, Austin et al. 2018). The typical concern is that in-migration dilutes the benefits of local investment. But schools build human capital that students carry with them, and so out-migration can magnify the gains for people while distributing them across locations. This perspective connects to work on education and economic geography (Eckert and Kleineberg 2024, Agostinelli et al. 2026, Fabre 2026, Desmet et al. 2026, Milsom 2026). Relative to these papers, I estimate the model with quasi-experimental variation from a national school construction program, and I use the model to study spatial incidence and policy design. Migration affects both who gains from school construction and where schools should be built.

Methodologically, I build most directly on Duflo (2001) and Bryan and Morten (2019). The former introduces the *Inpres* variation, and the latter quantifies migration frictions with a spatial equilibrium model.[1] In combining the two, my approach relates to a broader theme in empirical work. Well-identified effects can discipline economic models, and economic models can broaden the external validity of these effects (Nakamura and Steinsson 2018). Relative to Bryan and Morten (2019), I endogenize schooling and use a natural experiment to estimate the model. Relative to Duflo (2001), I use the model to trace the gains from school construction across people and places. I quantify these gains for one of the largest school construction programs in history.

2 Model

Human capital is portable. In the model, individuals invest in schooling and migrate for work. School construction encourages education, and migration distributes the gains across labor markets. Appendix A provides derivations.

2.1 Education and migration

Building on Bryan and Morten (2019) and Hsieh et al. (2019), individuals make schooling and migration choices in two stages of life: school and work.[2] First, an individual i born in origin j and cohort k chooses schooling. Second, they realize productivity draws across locations and choose where to work.[3] These choices determine the supply of human capital across labor markets. Schooling e produces human

[1] Further work on the *Inpres* program includes Duflo (2004), Martinez-Bravo (2017), Mazumder et al. (2019), Ashraf et al. (2020), Akresh et al. (2023), and Bazzi et al. (2026).

[2] Relative to Bryan and Morten (2019), I endogenize schooling and allow it to lower migration frictions. Relative to Hsieh et al. (2019), individuals choose schooling before their destination-specific draws are realized, and so human capital is not tailored to a particular destination. The model also relates to a broader literature on self-selection in labor markets (Roy 1951, Heckman 1974, Heckman and Sedlacek 1985, Keane and Wolpin 1997) and migration across labor markets (Dahl 2002, Kennan and Walker 2011, Moretti 2011, Young 2013).

[3] I abstract from migration during childhood, which can alter exposure to schools and neighborhoods (Chetty et al. 2016, Chyn and Katz 2021, Laliberté 2021). I also abstract from child labor, which reflects local labor demand and household circumstances (Atkin 2016, Shah and Steinberg 2017, Bau et al. 2026). Heterogeneous skills and schooling costs can accommodate these differences in childhood environments.

capital as follows.

$$h_i(e) = a_i s_{jk} \exp(\rho e) \tag{1}$$

Parameter $\rho > 0$ governs how much schooling raises human capital, subject to individual ability a_i and origin-cohort skill s_{jk}. I solve the model backward.

Migration

In the second stage, individuals consider locations ℓ. They take schooling e as given, and they realize productivity draws $\varepsilon_{i\ell}$ across locations. They choose a location to maximize utility.

$$\ell_i = \arg\max_{\ell} u_{i\ell}, \quad u_{i\ell} = \frac{w_{i\ell}}{\tau_{jk\ell}(e)} \tag{2}$$

Utility $u_{i\ell}$ depends on wages $w_{i\ell}$ and migration frictions $\tau_{jk\ell}(e)$. Wages compensate human capital at wage rates $r_{k\ell}$, which vary across cohorts and locations.

$$w_{i\ell} = r_{k\ell} h_i(e) \varepsilon_{i\ell} \tag{3}$$

Migration frictions are the financial, psychological, informational, and opportunity costs of moving and working away from home, net of destination amenities.

$$\tau_{jk\ell}(e)^{\theta} = \tau^0_{jk\ell} \exp(-\lambda e) \quad \text{for} \quad \ell \neq j, \tag{4}$$

and $\tau_{jkj}(e) = 1$ at home. Mobility parameter $\lambda \geq 0$ governs how much schooling reduces migration frictions relative to baseline levels $\tau^0_{jk\ell}$.

I characterize migration choices and expected utility. As in Eaton and Kortum (2002), productivity draws $\varepsilon_{i\ell}$ follow a Fréchet distribution with shape parameter $\theta > 1$ and mean one, where a high value of θ implies low dispersion. These draws are independent across locations and independent of ability, costs, and schooling. The Fréchet distribution delivers closed-form expressions. It is convenient to define access Ω_{jk} as a friction-adjusted measure of proximity to high wage rates.

$$\Omega_{jk}(e) = \sum_{\ell} \left(\frac{r_{k\ell}}{\tau_{jk\ell}(e)} \right)^{\theta} \tag{5}$$

Migration choices follow from equation 2. The probability of choosing location ℓ is

$$\pi_{jk\ell}(e) = \Pr(\ell_i = \ell) = \frac{\left(r_{k\ell}/\tau_{jk\ell}(e)\right)^{\theta}}{\Omega_{jk}(e)}. \tag{6}$$

This probability rises with the local wage rate, net of migration frictions. It falls with access, as other locations pull individuals away. Human capital $h_i(e)$ is common across locations and thus does not directly affect location choice. The probability of leaving home is

$$m_{jk}(e) = 1 - \pi_{jkj}(e) = 1 - \frac{r_{kj}^{\theta}}{\Omega_{jk}(e)}, \tag{7}$$

and the expected utility per unit of human capital is

$$\bar{r}_{jk}(e) = \frac{1}{h_i(e)}\,\mathbb{E}\Big[\max_{\ell} u_{i\ell}\Big] = \Omega_{jk}(e)^{1/\theta}. \tag{8}$$

The expectation is over productivity draws. Expected utility reflects access that rises with wage rates and falls with migration frictions. Without migration, $m_{jk}(e) = 0$ implies $\bar{r}_{jk}(e) = r_{kj}$ by equation 7. Expected utility is simply the home wage rate. With migration, $m_{jk}(e) > 0$ implies $\bar{r}_{jk}(e) > r_{kj}$. Expected utility exceeds the home wage rate, as migration creates option value that raises the value of human capital.

Education

In the first stage, individuals invest in schooling in anticipation of future migration and work. They choose schooling to maximize expected utility net of schooling costs.

$$e_i = \arg\max_{e \geq 0} \left\{ \bar{r}_{jk}(e) h_i(e) - \frac{1}{2} c_i h_i(e)^2 \right\}, \tag{9}$$

where expected utility $\bar{r}_{jk}(e)$ is given by equation 8. Schooling costs are quadratic in human capital, subject to individual cost shifter c_i. Ability and costs may be correlated.[4] For an interior optimum, the first-order condition implies that schooling e_i satisfies

$$\log h_i(e_i) = \underbrace{-\log c_i}_{\text{costs}} + \underbrace{\log r_{kj}}_{\text{home}} + \underbrace{\log \Lambda_{jk}(e_i)}_{\text{away}}. \tag{10}$$

[4] Costs determine how much human capital an individual acquires, and ability determines how many years of schooling it takes.

Equation 10 separates three forces. First, schooling *costs* c_i directly reduce schooling. Second, *home* wage rate r_{kj} encourages schooling by raising the value of human capital at home. Third, the *away* term captures the spatial determinants of schooling.

Space enters the away term through the probability $m_{jk}(e)$ of leaving home, which rises with access to high wage rates elsewhere.

$$\log \Lambda_{jk}(e) = -\frac{1}{\theta} \log \left[1 - m_{jk}(e)\right] + \log \left[1 + \frac{\lambda}{\rho\theta} m_{jk}(e)\right] \geq 0 \tag{11}$$

When no one leaves home, $\log \Lambda_{jk}(e) = 0$ and schooling has no spatial component. Otherwise, the away term is positive, and it rises with the probability of leaving home. The first term captures the option value of migration, as access to opportunities across labor markets raises the value of schooling – even for those who eventually choose to stay at home. The second term captures that schooling itself increases mobility when $\lambda > 0$. Schooling then raises the option value of migration, which further increases the return to schooling. Thus, education and migration are complements: schooling facilitates migration, and migration opportunities encourage schooling.

2.2 Production

Firms use human capital H_ℓ to produce output Y_ℓ in each location, subject to local productivity A_ℓ and output elasticity κ.

$$Y_\ell = A_\ell H_\ell^\kappa \tag{12}$$

The productivity term captures spatial heterogeneity in the productive use of human capital, including through differences in technology, physical capital, and sectoral composition. The output elasticity captures the net effect of local spillovers. Congestion reflects competition for land, capital, and customers, while agglomeration reflects innovation and economies of scale. Congestion dominates when $\kappa < 1$, and agglomeration dominates when $\kappa > 1$. Human capital in a location ℓ aggregates over cohorts k with Cobb–Douglas weights $\alpha_{k\ell}$.

$$H_\ell = \prod_k \left(H_{k\ell}\right)^{\alpha_{k\ell}}, \quad \alpha_{k\ell} > 0, \quad \sum_k \alpha_{k\ell} = 1 \tag{13}$$

Firms take local productivity as given and compete for human capital. Within a cohort and location, each unit of human capital $H_{k\ell}$ earns a common wage rate $r_{k\ell}$. Cobb–Douglas production implies that each cohort receives share $\alpha_{k\ell}$ of local output Y_ℓ. The wage bill is given by $r_{k\ell}H_{k\ell} = \alpha_{k\ell}Y_\ell$, and the wage rate follows directly.

$$r_{k\ell} = \alpha_{k\ell}\frac{Y_\ell}{H_{k\ell}} = \alpha_{k\ell}A_\ell\frac{H_\ell^\kappa}{H_{k\ell}} \tag{14}$$

Aggregate output sums across locations.

$$Y = \sum_\ell Y_\ell = \sum_\ell W_\ell, \quad W_\ell = \sum_k r_{k\ell}H_{k\ell} \tag{15}$$

2.3 Labor markets

Individuals supply labor where they work, firms demand labor for production, and wage rates clear labor markets in each location. Each labor market pools human capital from many origins.

Supply

Individuals supply effective labor $h_i\varepsilon_{i\ell}$ in their chosen locations, where $h_i = h_i(e_i)$ at chosen schooling e_i. Observed wages reflect selection: those who choose location ℓ have more favorable draws $\varepsilon_{i\ell}$, especially for rarely chosen locations. The Fréchet distribution characterizes this selection in terms of choice probabilities $\pi_{i\ell} = \pi_{jk\ell}(e_i)$.

$$\mathbb{E}\big[\varepsilon_{i\ell} \,|\, \ell_i = \ell\big] = \pi_{i\ell}^{-1/\theta} \tag{16}$$

Let $\mathcal{I}_k$ denote the individuals in cohort k. Labor supply in location ℓ is

$$H_{k\ell}^S(r) = \sum_{i\in\mathcal{I}_k} \pi_{i\ell}\,\mathbb{E}\big[h_i\varepsilon_{i\ell} \,|\, \ell_i = \ell\big] = \sum_{i\in\mathcal{I}_k} h_i\pi_{i\ell}^{1-1/\theta}. \tag{17}$$

Wage rates r enter through schooling and migration choices. Holding schooling fixed, supply is upward-sloping: a higher wage rate $r_{k\ell}$ raises choice probability $\pi_{i\ell}$ by attracting migrants. When schooling adjusts, wage rates also affect human capital through the schooling decision. By the law of large numbers, expected and realized supply coincide for a continuum of individuals, and idiosyncratic draws average out.

Demand

Firms demand local labor for production. For $\kappa \neq 1$, labor demand follows from inverting equation 14 in each location.

$$H^D_{k\ell}(r_\ell) = \frac{\alpha_{k\ell} R_\ell}{r_{k\ell}} \left(\frac{A_\ell}{R_\ell} \right)^{\frac{1}{1-\kappa}}, \quad R_\ell = \prod_k \left(\frac{r_{k\ell}}{\alpha_{k\ell}} \right)^{\alpha_{k\ell}} \tag{18}$$

Productivity A_ℓ shifts demand, and output elasticity κ governs its response. Local wage rates $r_\ell = \{r_{k\ell}\}_k$ determine both the cohort composition and aggregate scale of demand. Relative wage rates affect cohort composition, as an increase in a cohort's relative wage rate reduces its relative demand.[5] Wage index R_ℓ affects aggregate scale. For $\kappa < 1$, demand is downward-sloping: congestion lowers productivity as human capital grows, and so firms absorb more human capital only at lower wages. For $\kappa > 1$, demand is upward-sloping: agglomeration raises productivity as human capital grows, and so firms absorb more at higher wages. For $\kappa = 1$, demand is perfectly elastic at wage index $R_\ell = A_\ell$, although wage rates $r_{k\ell}$ can still adjust through changes in cohort composition.

Equilibrium

Equilibrium wage rates $r = \{r_{k\ell}\}_{k\ell}$ clear labor markets for each cohort k and location ℓ.

$$H^D_{k\ell}(r_\ell) = H^S_{k\ell}(r) \quad \forall\, k, \ell \tag{19}$$

Demand can be evaluated separately for each location because production is local. Supply must be evaluated jointly across locations because migration is national. Schooling shifts labor supply, and wage rates adjust in response.

2.4 Investment

Public investment operates through individual investment. School construction in an origin j reduces schooling costs for cohorts k that are young enough to benefit.

$$\log c_i = \log c_i^0 - \delta S_j T_k \tag{20}$$

[5] For two cohorts k and k' in a given location, equation 18 yields $H^D_{k\ell}/H^D_{k'\ell} = (\alpha_{k\ell}/\alpha_{k'\ell})(r_{k'\ell}/r_{k\ell})$.

Cost parameter $\delta \geq 0$ governs how much school construction S_j reduces schooling costs c_i for treated cohorts, as indicated by exposure T_k. In turn, these cost reductions shape individual decisions and outcomes. By equation 10, the direct effect of an additional unit of school construction $S_j T_k$ is to increase schooling by δ/ρ years. More broadly, consider the effects on expected wages

$$\bar{w}_i = \sum_\ell \pi_{i\ell}\, \mathbb{E}\big[w_{i\ell} \,|\, \ell_i = \ell\big] = \sum_\ell r_{k\ell} h_i \pi_{i\ell}^{1-1/\theta}, \tag{21}$$

which reflect human capital h_i, choice probabilities $\pi_{i\ell}$, and wage rates $r_{k\ell}$. Public investment affects the first two jointly because schooling and migration are intertwined. Holding wage rates fixed, I distinguish three channels.

$$\frac{\partial \log \bar{w}_i}{\partial S_j T_k} = \frac{\delta}{\rho} \Big(\underbrace{\rho}_{\text{Mincer}} + \underbrace{\mu_i}_{\text{mobility}} \Big) \cdot \underbrace{M_i}_{\text{motives}} \tag{22}$$

The direct schooling response δ/ρ translates construction into years of schooling. The ρ and μ_i terms are returns per year of schooling, and M_i is a multiplier.

$$\mu_i = \lambda \left(1 - \frac{1}{\theta}\right)(\tilde{m}_i - m_i), \quad M_i = M_{jk}(e_i) = \frac{\rho}{\rho - d \log \Lambda_{jk}(e_i)/de}, \tag{23}$$

where $\tilde{m}_i = \sum_{\ell \neq j} \pi_{i\ell}\, \mathbb{E}[w_{i\ell} \,|\, \ell_i = \ell]/\bar{w}_i$ is the expected share of wages earned away from home.

Mechanisms

First, the *Mincerian* return captures the direct effect of schooling on wages. In the first stage, holding migration and wage rates fixed, a unit of public investment $S_j T_k$ increases schooling by δ/ρ years. By equations 1 and 3, human capital and wages then rise by ρ log points per year.

Second, the *mobility* return captures that migration responds to schooling. In the second stage, holding schooling and wage rates fixed, individuals take their human capital where wages are highest net of migration frictions, which keep some of them from where they would earn the most. For $\lambda > 0$, the additional schooling lowers these frictions by equation 4, increasing access to high wages and drawing more individuals away from home. The mobility return is positive when migration frictions are costs

rather than amenities on net.

Third, the *motives* multiplier allows schooling to respond in turn. In the first stage, holding wage rates fixed, individuals anticipate the option value of future migration, which rises with access to a wide set of work opportunities. By equation 8, this access raises the value of human capital and thus encourages schooling. For $\lambda > 0$, schooling encourages migration by lowering frictions, and migration encourages schooling by widening access. This feedback amplifies the direct effect on years of schooling, and each additional year earns both the Mincerian and mobility returns.

Finally, the *market* channel allows wage rates to adjust. Public investment induces schooling and migration responses, which shift the distribution and local composition of labor supply. Wage rates move accordingly to clear labor markets, and these equilibrium adjustments spill over to untargeted origins and cohorts.

Incidence

Public investment pays off for both people and places. For people, I compute the total wage gains from investment in an origin j by summing over the individuals born in j. Expected wages $\bar{w}_i$ are given by equation 21, and j_i denotes the origin of individual i.

$$\Delta \bar{W}_j = \sum_i \Delta \bar{w}_i \cdot \mathbb{1}\{j_i = j\} \tag{24}$$

For places, I compute wage gains in the locations ℓ where individuals work. These gains are given by the wage bill of equation 15, which pools individuals across origins.

$$\Delta W_\ell = \Delta \left(\sum_k r_{k\ell} H_{k\ell} \right) \tag{25}$$

Gains for people and places differ because of migration. Summing across origins and destinations, both yield the same aggregate gain.

The channels of interest determine where these gains accrue, even as public investment targets specific origins. Gains for places reach farther than gains for people. At fixed wage rates, gains for people appear only at these origins, while gains for places appear wherever people from these origins choose to work. The Mincerian return accrues wherever they work, and so the gains reach other locations even with migration patterns held fixed. The mobility return draws more of them away from

home, bringing these gains even farther. The motives multiplier scales both. The exception is the market channel, which widens the reach of gains for people. Wage rates adjust in all places where labor supply shifts, and these adjustments reach people from untargeted origins.

3 Setting

Indonesia's *Sekolah Dasar Inpres* program built primary schools at national scale in the 1970s. I describe the program and data. Appendix B provides additional detail on data sources and construction.

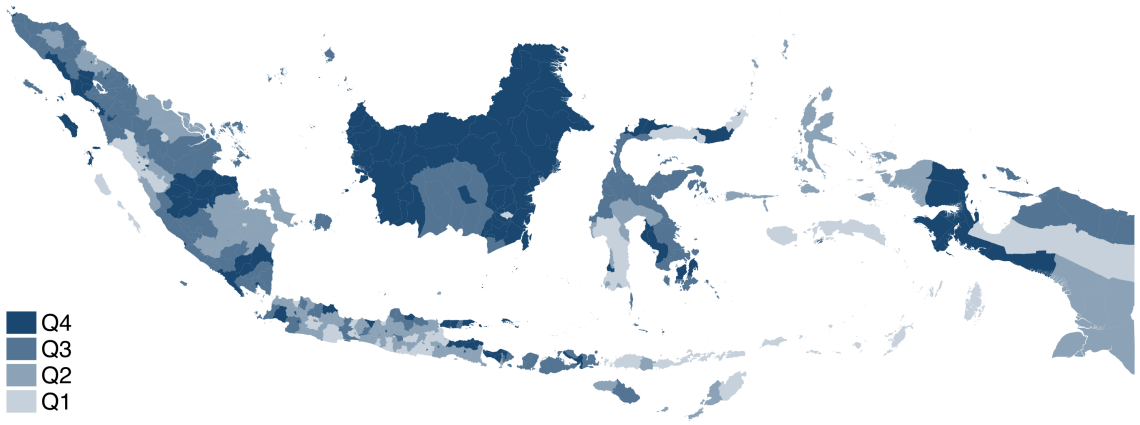
3.1 Program

Inpres refers to the presidential instructions (*Instruksi Presiden*) that established the *Inpres* program across two five-year development plans (*Repelita*). *Repelita I* focused on infrastructure investment from 1969 to 1974, and *Repelita II* placed additional priority on education from 1974 to 1979. Coupled with windfall oil revenues, these government priorities enabled school construction of unprecedented scale. The program aimed to build 62,000 new primary schools nationwide: 6,000 per year in 1973 and 1974, 10,000 per year in 1975 and 1976, and 15,000 per year in 1977 and 1978. The program also funded teacher recruitment and compensation, as well as concurrent improvements in water and sanitation systems.[6]

Figure 1 maps the allocation of *Inpres* school construction across districts. The program targeted low-enrollment districts. In 1973 and 1974, school construction was allocated across districts in proportion to the pre-program number of unenrolled children of primary school age. From 1975 to 1978, unenrollment was defined relative to a 15% threshold. Districts with unenrollment rates below 15% received no new schools, while those above 15% received new schools in proportion to their child population times the amount by which their pre-program unenrollment rate exceeded 15%. Appendix figure B1 shows that school construction across districts follows these population-weighted allocation measures in the data. "Districts" refers collectively

[6] *Inpres* No. 10/1973, 6/1974, 6/1975, 3/1976, 3/1977, and 6/1978 document these plans. Large-scale infrastructure investment is typical of Indonesian development policy, with similar efforts for roads (Gertler et al. 2024), healthcare (Hsiao 2022), and flood protection (Hsiao 2024, 2025).

Figure 1: *Inpres* school construction



Each observation is a district. I plot quartiles of the number of *Inpres* schools constructed from 1973 to 1978 per 1,000 children in 1971. Children are those of age 5 to 14.

to regencies (*kabupaten*) and municipalities (*kota*), each of which is a subprovincial administrative unit.

Rural development and regional equity were among the program's stated goals.[7] Rural development calls for construction in rural districts, and regional equity calls for the benefits of development to be spread across districts. While the program concentrated construction in specific districts, the benefits may have spread through migration. The spatial distribution of benefits depends on where people choose to work. Evaluating the program's policy goals thus requires distinguishing gains for people from gains for places, and tracing both to the districts that received schools. The model of section 2 provides exactly this accounting, and the program provides the variation to estimate it.

3.2 Treatment data

District-level data on *Inpres* school construction come from Duflo (2001), which draws on Ministry of National Development Planning (*Bappenas*) reports from 1973 to 1978 and population census reports from 1971. I observe the number of *Inpres* primary schools constructed from 1973 to 1978, *Inpres* water and sanitation spending

[7] *Inpres* No. 6/1974 calls for "expanding learning opportunities [...] especially in rural areas" ("*memperluas kesempatan belajar [...] terutama di daerah pedesaan*"). The 1979/80 budget states the goal of "evenly distributing the benefits of development across all regions" ("*agar hasil-hasil pembangunan dapat merata ke seluruh daerah*," Republik Indonesia 1979).

Table 1: Treatment data

	Mean	SD	Min	Max	N
Inpres schools per 1,000 children	2.26	1.09	0.69	8.29	282
Inpres school construction	220	175	16.0	824	282
Population (1000s)	423	362	12.4	1,975	282
Child population (1000s)	115	98.0	3.80	543	282
Child enrollment rates	0.51	0.14	0.02	0.84	282
Log population density	5.05	1.96	0.04	10.1	282

Each observation is a district. Columns show means, standard deviations, minima, maxima, and the number of observations. Rows summarize the number of *Inpres* schools constructed from 1973 to 1978 and population, enrollment, and population density in 1971. Children are those of age 5 to 14.

from 1973 to 1978, child populations in 1971, and child enrollment rates in 1971. The data record planned school construction, which Duflo (2001) argues aligns well with actual school construction.

I verify the data by consulting the original primary sources. For school construction, I use the *Bappenas* reports to correct a modest number of inconsistencies. For child populations, I transcribe the census reports and again make limited corrections. Children are defined as those of age 5 to 14. For child enrollment, I construct child enrollment rates with a random sample of the census microdata. I draw total populations from the census reports, and I take the water and sanitation spending data as given in Duflo (2001). Appendix table B2 compares the resulting data to those in Duflo (2001) and shows that the data remain similar.

Table 1 summarizes the data. The average district received 220 new primary schools from the *Inpres* program, yielding 2.26 new primary schools per 1,000 children recorded in 1971. This measure is the main treatment variable. Child enrollment rates in 1971 were low at 51% on average and did not exceed 84%. For the district with the lowest level of child enrollment, only 2% of school-aged children attended school. I measure ruralness with population density in 1971, defined as total population per square kilometer of land area. Appendix figure B1 shows that the program built more schools per child in low-density districts, in line with its goal of rural development.

3.3 Outcome data

Individual-level data on socioeconomic outcomes come from the 2011, 2012, 2013, and 2014 National Socioeconomic Surveys (*Susenas*). That is, I measure long-run outcomes four decades after the first *Inpres* schools were completed in 1974. Following Duflo (2001), I restrict attention to male individuals of age 2 to 24 in 1974. Individuals of age 2 to 6 in 1974 were young enough to attend the new primary schools, while those of age 12 to 17 were too old. I refer to these groups as the exposed and unexposed cohorts. I observe districts of residence and birth, which I harmonize to 1973 boundaries, and I use birth districts to determine exposure to *Inpres* school construction. The assumption is that individuals pursue primary schooling where they are born. I define migrants as those who reside outside of their birth districts.[8]

The data record education. I observe the highest level of schooling attended and the number of years within each level. I compute the years of schooling completed, counting primary school as six years, middle school as three, high school as three, community college as two, and university as four. I pool vocational, religious, and traditional schooling for middle school and high school. I subtract one from the number of years attended if individuals report not having completed a given level of schooling. Tertiary schooling is subject to measurement error: community college programs range from one to three years, and master's and doctoral programs exceed four years. I take twelve as the maximum years of schooling to avoid this measurement issue, given that the impacts of *Inpres* primary schools likely concentrate at lower levels.

The data also record wages. I measure monthly net wages from an individual's main job, inclusive of both money and goods, as well as hours worked in the past week. I use these measures to compute log hourly wages. I define the employed as those who report working as their primary activity in the past week. I observe whether individuals work as employees, employers, own-account workers, or unpaid family workers. I code employees as wage employed. The data measure income for some of the non-wage employed, but appendix table B3 shows these data to be incomplete. I therefore focus on wages in my analysis. Duflo (2001) discusses the

[8] International out-migration is small. Net out-migration was 905,000 from 1980 to 2015, relative to a total population of 259 million in 2015 (World Bank 2022).

Table 2: Outcome data

	Mean	SD	N
Years of schooling	7.30	3.92	483,503
Log hourly wages	9.12	0.88	164,046
Migration	0.25	0.43	483,503
Completed primary school	0.75	0.43	483,503
Completed middle school	0.44	0.50	483,503
Completed high school	0.31	0.46	483,503
Employed	0.91	0.29	483,503
Wage employed	0.41	0.49	438,909

Each observation is a male individual of age 2 to 24 in 1974. Columns show means, standard deviations, and the number of observations. Rows summarize schooling, wages, and migration. Schooling is years completed. Wages are log hourly net wages in year-2011 Indonesian rupiah from an individual's main job. Migrants are those whose districts of birth and residence do not coincide. The employed are those who report working as their primary activity. Among the employed, the wage employed are employees rather than employers, own-account workers, or unpaid family workers. Districts are based on 1973 boundaries.

potential selection issue at length. I also construct a district-level wage panel from the 1986 to 1997 National Labor Force Surveys (*Sakernas*), which measure local wages as exposed cohorts entered the labor market.

Table 2 summarizes the data. Average schooling is 7.30 years, and average log hourly wages are 9.12 in year-2011 rupiah – roughly US$1 per hour. There is meaningful migration at 25% of individuals, with 16% living outside their province of birth. This mobility creates scope for investments in one district to affect labor markets and outcomes in other districts. The completion rate is 75% for primary school, but falls to 44% for middle school and 31% for high school. Employment is high at 91%, although only 41% of the employed are wage workers.

4 Estimation

I estimate the model by exploiting the identifying variation of the *Inpres* program, and I present the resulting estimates. Appendix C details the estimation procedure and provides additional results.

4.1 Identification

I estimate parameters $(\delta, \rho, \lambda, \kappa, \theta)$. For (δ, ρ, λ), I draw directly on the *Inpres* program. As in Duflo (2001), I use differences in program exposure across districts and cohorts to estimate the causal effects of school construction on schooling, wages, and migration. These effects jointly identify the parameters that govern schooling costs, human capital returns, and migration frictions. I use indirect inference to recover parameters that exactly reproduce the causal estimates. For κ, I again draw on the *Inpres* program. As in Duflo (2004), I follow unexposed cohorts as exposed cohorts enter the labor market to estimate equilibrium wage effects. I use the model to interpret these effects, which discipline the balance between congestion and agglomeration. For θ, I compare wages and location shares across destinations. The model links this relationship to the strength of selection.

4.2 Schooling, wages, and migration

I compare exposed and unexposed cohorts in districts with high and low levels of school construction. I define these cohorts by age: individuals of age 2 to 6 in 1974 were young enough to attend *Inpres* schools, while those of age 12 to 17 were too old. Cohorts are thus analogous to time in the typical difference-in-differences setting. Construction intensity S_j records the number of *Inpres* schools constructed from 1973 to 1978 per 1,000 children in 1971. For individual i born in origin j and birth year k, observed in survey year t, I estimate the specification

$$y_i = \alpha_j + \alpha_k + \alpha_t + \beta S_j T_k + T_k C_j' \gamma + \varepsilon_i. \tag{26}$$

Coefficient β captures the effect of construction intensity on exposed cohorts, as indicated by exposure T_k. Fixed effects α control for birth district j, birth year k, and survey year t, and standard errors are clustered by birth district.

The identifying assumption is parallel trends in high- and low-construction districts absent the program. Controls C_j, which I interact with exposure, address the main threats to this assumption. First, schools were allocated as a function of child population and enrollment in 1971, and so high- and low-construction districts differed by these measures to begin with. Second, the *Inpres* program included concurrent

water and sanitation spending, the effects of which should not be attributed to school construction. Third, cohort trends may have differed by population size and island group. In particular, Indonesian regions followed distinct cohort trajectories over this period, driven by resource development and transmigration in the outer islands and rapid urbanization in Java. I control for each of these potential confounders.[9] I test for parallel trends with a placebo comparison of individuals of age 12 to 17 and 18 to 24 in 1974, none of whom was exposed to the program.

The program raises schooling, wages, and migration, as measured four decades after school construction. Table 3a shows that each additional school per 1,000 children raises schooling by 0.148 years. Wages rise by 0.031 log points, and migration rises by 1.23 percentage points. None of the placebo estimates is statistically distinguishable from zero. Tables 3b and 3c use the induced schooling to estimate its effects on wages and migration. I instrument for years of schooling with $S_j T_k$ and report Anderson–Rubin (AR) confidence sets, which are robust to weak instruments (Anderson and Rubin 1949). Each induced year of schooling raises wages by 0.213 log points and migration by 8.34 percentage points. I estimate long-run wage effects that are roughly twice as large as the medium-run estimates of Duflo (2001), as well as the global average reported by Psacharopoulos and Patrinos (2018). Larger long-run effects may reflect rising returns to schooling over the life cycle.[10]

I recover model parameters by indirect inference, as detailed in appendix C, taking columns (1), (8), and (10) of table 3 as target moments. With three model parameters, I match the three regression coefficients exactly. The regression coefficients cannot be read directly as model parameters for two reasons. First, the regressions measure the combined effects of multiple channels rather than isolating individual parameters. For example, the wage return in table 3b includes changes in migration choices and equilibrium wage rates, while parameter ρ measures the human capital return holding these margins fixed. Second, the regressions aggregate responses across treatment intensities with weights determined by the distribution of intensity and the

[9] Duflo (2001) includes the first two sets of controls. Appendix table C1 replicates this analysis with and without the third set of controls, which modestly increase the estimated magnitudes.

[10] My IV estimate of the wage effect also exceeds its OLS counterpart, consistent with the broad pattern documented by Gethin (2025). Measurement error in reported schooling may attenuate OLS, particularly because the *Susenas* surveys rely on recall many years after school construction. For migration, employment opportunities at home may encourage schooling while discouraging departure, further lowering OLS relative to IV.

Table 3: Schooling, wage, and migration effects

(a) School construction

	Treatment			Placebo		
	Years of schooling	Log wages	Migration	Years of schooling	Log wages	Migration
	(1)	(2)	(3)	(4)	(5)	(6)
Inpres effect	0.148***	0.031***	1.23**	0.004	−0.018	−0.31
	(0.042)	(0.009)	(0.54)	(0.060)	(0.011)	(0.62)
Observations	98,413	98,413	98,413	59,727	59,727	59,727
Control mean	8.43	9.19	33.1	7.05	8.89	28.3
Implied δ	0.03					

(b) Log wages

	OLS	IV
	(7)	(8)
Years of schooling	0.109***	0.213***
	(0.002)	(0.056)
AR 95% set		[0.11, 0.39]
First-stage F		12.2
Observations	98,413	98,413
Control mean	9.19	9.19
Implied ρ		0.19

(c) Migration

	OLS	IV
	(9)	(10)
Years of schooling	2.06***	8.34**
	(0.11)	(4.10)
AR 95% set		[1.3, 21.7]
First-stage F		12.2
Observations	98,413	98,413
Control mean	33.1	33.1
Implied λ		0.44

Each column is a regression, and each observation is a male wage earner. Individuals are of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974, except that the placebo columns compare individuals of age 12 to 17 and 18 to 24. Schooling is years completed, wages are log hourly wages, and migration is residence outside of the birth district in percentage points. Panel (a) regresses each outcome on $S_j T_k$, and panels (b) and (c) regress wages and migration on schooling. I include fixed effects for birth district, birth year, and survey year, as well as controls for child population and enrollment, *Inpres* water and sanitation spending, population decile, and island group. I interact all controls with exposure T_k, and I cluster standard errors by birth district. The IV columns instrument for schooling with $S_j T_k$ and report Anderson–Rubin confidence sets. Control means refer to unexposed cohorts. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

regression controls. This aggregation complicates interpretation when responses differ across intensities (Callaway et al. 2026).[11] Appendix table C2 reports these weights, which include negative values, and provides suggestive evidence of heterogeneity in schooling effects, although homogeneity is not rejected for wages or migration. Indirect inference provides a model-consistent mapping from coefficients to parameters by applying the same regressions to model-implied responses.

4.3 Equilibrium

I compare unexposed cohorts before and after exposed cohorts enter the labor force in districts with high and low levels of school construction. As exposed cohorts reach working age, districts with more school construction experience larger increases in the supply of educated labor. I use this labor supply shock to study equilibrium wage effects. School construction does not directly affect unexposed cohorts, and so their wage responses capture equilibrium adjustments. By contrast, the cohort comparison of section 4.2 has a "missing intercept" problem: it evaluates exposed cohorts only relative to unexposed cohorts from the same district. It cannot identify equilibrium effects in levels because wages adjust for both cohorts.

Equilibrium wage effects combine two forces. On one hand, an increase in aggregate labor supply affects wages through congestion and agglomeration. On the other hand, when cohorts are imperfect substitutes, an increase in labor supply for exposed cohorts raises relative wages for unexposed cohorts, whose labor becomes relatively scarcer. I use the model to separate these forces by adjusting wages for relative labor supply. For district ℓ in survey year t, I estimate the specification

$$\log w_{U\ell t} + \log \frac{H_{U\ell t}}{H_{\ell t}} = \alpha_\ell + \alpha_t + (\kappa - 1) \log H_{\ell t} + C'_\ell \gamma_t + \varepsilon_{\ell t}, \tag{27}$$

where $w_{U\ell t}$ is wages among unexposed cohorts, $H_{U\ell t}$ is their human capital, and $H_{\ell t}$ is aggregate human capital. The human capital terms measure the total supply of productive labor. Appendix C derives the specification from the model and details the construction of these terms.

I use data from the 1986 to 1997 *Sakernas* labor force surveys, restricting at-

[11] I use a common definition of cohort exposure across districts, relying on variation in treatment intensity rather than staggered adoption.

tention to male individuals of age 20 to 60. Unexposed cohorts are those born by 1962. I measure labor supply for each cohort as the years of schooling embodied in wage employment, and I aggregate across cohorts with equation 13. In estimating equation 27, an endogeneity issue arises because local productivity shocks may affect both labor supply and wages. The *Inpres* program provides a shifter: I instrument for aggregate labor supply with $S_\ell E_t$, where S_ℓ is school construction intensity and E_t indicates survey years from 1992, when the youngest exposed cohort reaches age 20 and enters the panel. Fixed effects α control for district ℓ and survey year t, and standard errors are clustered by district. Controls C_ℓ are those of equation 26 with year-specific coefficients. The identifying assumption is again parallel trends in high- and low-construction districts, now absent the entry of exposed cohorts. I assess this assumption with year-by-year estimates, which test for differential trends before 1992.

The *Inpres* variation points to net congestion, although the estimates are imprecise. Table 4a reports the effects of school construction on labor supply and wages, adjusting for relative labor supply. Each additional school per 1,000 children raises labor supply by 0.055 log points as exposed cohorts enter the labor market, while adjusted wages change by -0.008 log points. Table 4b uses the induced increase in labor supply to estimate a wage elasticity. The OLS estimate is positive, consistent with endogeneity bias from productivity shocks that raise both labor supply and wages. The IV estimate is -0.153 and corresponds to $\kappa - 1$, which yields $\kappa = 0.85$. The point estimate for κ is less than one, indicating that congestion outweighs agglomeration, although the wide confidence set includes larger net effects in either direction.

I also examine the timing of effects and sensitivity to alternative specifications. Appendix figure C1 plots year-by-year estimates and shows no clear evidence of differential trends in labor supply or wages before 1992, when the youngest exposed cohort reaches working age. Appendix table C3 assesses sensitivity to the construction of human capital: using wage weights and exponential weights to measure cohort-specific human capital, and using annual shares and population scaling to estimate production weights for aggregating across cohorts. The estimates remain similar to baseline.

4.4 Selection

I compare wages across destinations among individuals with the same origin, cohort, and years of schooling. This comparison is informative about selection: individ-

Table 4: Equilibrium effects

(a) School construction

	Log labor supply	Log wages
	(1)	(2)
Inpres effect	0.055**	−0.008
	(0.022)	(0.013)
Observations	2,596	2,596
Control mean	11.58	8.59

(b) Log wages

	OLS	IV
	(3)	(4)
Log labor supply	0.157***	−0.153
	(0.023)	(0.255)
AR 95% set		[−1.19, 0.39]
First-stage F		6.4
Observations	2,596	2,596
Control mean	8.59	8.59
Implied κ		0.85

Each column is a regression, and each observation is a district-year cell of male individuals of age 20 to 60. Labor supply is log aggregate human capital, constructed from total years of schooling in wage employment by cohort. Wages are mean log hourly wages of unexposed cohorts born by 1962, adjusted for relative labor supply. Panel (a) regresses each outcome on $S_\ell E_t$, and panel (b) regresses wages on labor supply. I include fixed effects for district and survey year, as well as controls for child population and enrollment, *Inpres* water and sanitation spending, population decile, and island group. I interact all controls with survey year, and I cluster standard errors by district. The IV column instruments for labor supply with $S_\ell E_t$ and reports Anderson–Rubin confidence sets. Control means refer to survey years before 1992 with wages in 2011 prices. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

uals' productivity in each destination affects their location choices, but productivity is unobserved. In the model, individuals who choose less common destinations have more favorable productivity draws on average, and these favorable draws are reflected in higher wages. For individual i born in district j and cohort k, with schooling e_i and living in district ℓ in survey year t, I estimate the specification

$$\log w_i = \alpha_{jk}(e_i) + \alpha_{k\ell t} - \frac{1}{\theta} \log \hat{\pi}_{jk\ell t}(e_i) + \varepsilon_i, \tag{28}$$

where w_i is hourly wages and $\hat{\pi}_{jk\ell t}(e_i)$ is the estimated probability of choosing district ℓ. Fixed effects $\alpha_{jk}(e_i)$ control for differences in human capital by birth district, cohort, and schooling, while fixed effects $\alpha_{k\ell t}$ control for wage rates by cohort, destination, and survey year. Standard errors are clustered by birth district. The model implies that the coefficient on log location share equals $-1/\theta$, where θ is the Fréchet shape parameter. A higher value of θ implies weaker selection, less dispersion in destination productivity draws, and location choices that are more responsive to relative

Table 5: Selection effects

	Log wages	
	No fixed effects	Fixed effects
	(1)	(2)
Log location share	−0.119***	−0.044***
	(0.005)	(0.002)
Observations	97,165	97,165
Control mean	9.20	9.20
Implied θ		22.9

Each column is a regression, and each observation is a male wage earner of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974. Wages are log hourly wages. Location shares are calculated within groups defined by birth district, cohort, and schooling, pooling years of schooling into three bands and pooling across survey years. Column (2) includes fixed effects for birth district by cohort by years of schooling, as well as fixed effects for cohort by destination by survey year. I cluster standard errors by birth district. Control means refer to unexposed cohorts. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

wage rates. Appendix C derives equation 28 and describes how I estimate choice probabilities $\hat{\pi}_{jk\ell t}(e_i)$ from empirical location shares.

The estimates indicate positive selection into less common destinations. Table 5 reports the elasticity of wages with respect to location shares. I obtain an estimate of −0.119 without fixed effects and −0.044 with the fixed effects of equation 28. For a given origin, the fixed-effects estimate implies that a destination chosen by half as many individuals carries a wage premium of approximately 3%. Appendix table C4 reports four robustness checks: calculating location shares by birth year rather than cohort, by years of schooling rather than pooling into bands, by survey year rather than pooling across years, and excluding sparse cells with fewer than five observations. The estimates remain similar to baseline.

5 Counterfactuals

I quantify how mobility shapes the long-run aggregate and distributional effects of school construction, and I highlight the equity-efficiency trade-off facing the policymaker. Appendix D details how I solve the model and provides additional results.

5.1 Incidence

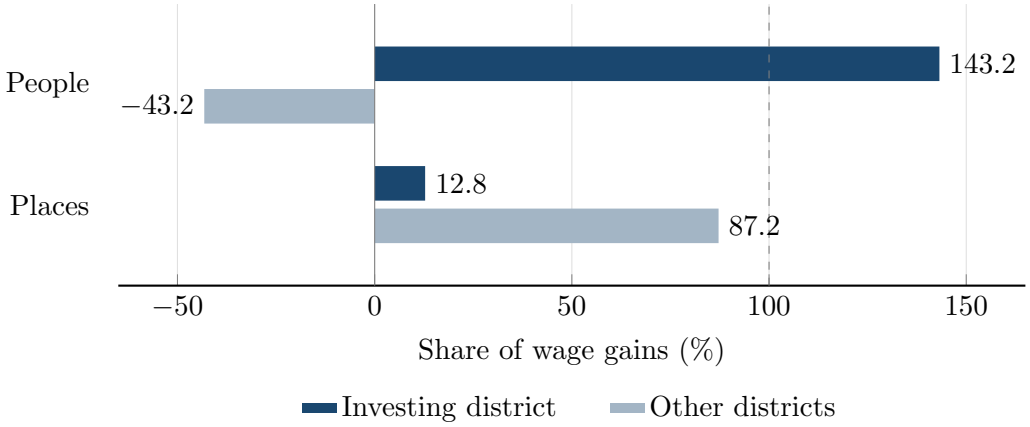
School construction targets specific districts, but the benefits may accrue elsewhere. I quantify the implications for spatial incidence by distinguishing gains for people born in a district from gains for places where people work. The first account follows individuals wherever they choose to work, while the second focuses on the economic production that occurs within each district. Both accounts sum to the same aggregate wage gains, which coincide with aggregate output gains by equation 15. I analyze spatial incidence by evaluating the impact of an additional school in each district. Starting from observed school construction, I add one school in a given district, and I solve for the resulting changes in schooling, migration, and wage rates. I repeat this exercise separately for each of the 282 districts.

Figure 2 reports the share of total gains, summing across simulations, that accrue to investing districts and to other districts. Investing districts are those that receive additional schools in each simulation, while other districts are those that do not. From the perspective of people, it is primarily those from investing districts who benefit from additional schools. These individuals experience 143.2% of total gains, while those from other districts experience losses equivalent to 43.2% of total gains. Migration and equilibrium adjustments drive these losses: as individuals from investing districts build human capital and move elsewhere, they increase the supply of productive labor in other districts and drive down wage rates in equilibrium. Lower wage rates hurt those from other districts.

By contrast, from the perspective of places, investing districts capture a small portion of the total gains from additional schools. Only 12.8% of wage gains are earned within investing districts, while 87.2% are earned in other districts. Many of those who benefit from additional schools eventually migrate elsewhere. These results suggest that school construction in a given district can generate substantial gains for people from that district, even if these gains do not remain within the district. Stated differently, concentrating school construction in high-return districts does not necessarily concentrate the economic gains within those districts. Furthermore, local wage gains are an incomplete measure of the benefits of school construction because migration generates large spillovers that distribute the gains across space.

At the same time, these results rely on the model's ability to predict migration

Figure 2: Incidence



I simulate the construction of one additional school in each of the 282 districts. For each simulation, I compute wage gains across people and places. Gains for people are those of people born in a district. Gains for places are those of people living in a district. The investing district is the district that receives the additional school. The x-axis reports the share of wage gains for each group, calculated as the sum of its wage gains across simulations divided by total wage gains across simulations. The sample is male wage earners of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974.

choices. Estimation directly targets the aggregate migration response to school construction, as estimated in table 3, but it does not target migration responses from particular origins or to particular destinations. Thus, as a validation exercise, I assess model fit for these untargeted moments. Appendix table D1 uses the *Inpres* variation to estimate migration responses from sparse and dense origins and to sparse and dense destinations. Population density serves as a measure of ruralness and thus proxies for access to large urban labor markets. The model reproduces the larger contribution from dense origins and the larger contribution to sparse destinations, providing some reassurance that it captures the underlying composition of migration responses.

5.2 Mechanisms

School construction builds human capital, but its effects depend on how individuals and labor markets respond. I use the model to distinguish four channels through which the *Inpres* program affects aggregate output and inequality. I begin by solving for a counterfactual baseline with no school construction. Next, I introduce observed school construction while holding migration and wage rates fixed at their baseline levels. I then allow migration to respond, followed by schooling feedback, and finally

Table 6: Mechanisms

	Output (%)	Inequality (%)	
		People	Places
Mincerian	3.18	−0.93	0.28
Mobility	0.34	−0.51	2.62
Motives	0.28	−0.04	0.37
Market	−1.21	0.29	−2.01
Total	2.58	−1.20	1.26

I evaluate the impacts of observed school construction relative to a baseline of no school construction. The first four rows each isolate the incremental impact of an economic channel. First, Mincerian introduces school construction while holding migration and wage rates fixed at their baseline levels. Second, mobility allows migration to respond to school construction. Third, motives allows migration and schooling to respond jointly. Fourth, market allows wage rates to adjust. The last row is the total effect of the program. I report impacts on output and inequality in percentage changes. Each channel's contribution is expressed relative to outcomes without construction, such that the first four rows sum to the last. Output is aggregate wages. Inequality is the population-weighted Gini coefficient of district mean wages. Inequality across people compares mean wages of people born in different districts. Inequality across places compares mean wages of people living in different districts. The sample is male wage earners of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974.

adjustment in wage rates. Each step contributes an incremental effect that isolates a particular channel of interest. These contributions sum to the total effect of the program.

Table 6 shows that *Inpres* school construction raised aggregate output while reducing inequality across people and increasing inequality across places. The last row reports the total effects: an output gain of 2.58%, a reduction in inequality across people of 1.20%, and an increase in inequality across places of 1.26%. I measure output with aggregate wages, as given by equation 15. I measure inequality with population-weighted Gini coefficients of district mean wages. Inequality across people compares mean wages of people born in different districts, while inequality across places compares mean wages of people living in different districts.[12]

The Mincerian channel captures the direct effects of school construction, holding migration and wage rates fixed. This channel reflects the conventional rationale

[12] The magnitude of the aggregate effects is consistent with the scale of the schooling response. The difference-in-differences estimate is 0.148 years of schooling for each additional school per 1,000 children, and the average construction intensity is 2.26 schools per 1,000 children. Multiplying gives 0.33 years of schooling, and equilibrium adjustments dampen the resulting output gains.

for school construction: lower costs of schooling increase individual investment in schooling, raising both human capital and wages. Output rises by 3.18% and inequality across people falls by 0.93%, while inequality across places rises by 0.28%. The effects on inequality across people and places diverge even for these direct gains.

The mobility channel captures the additional effects of migration responses, taking schooling as given and holding wage rates fixed. Schooling lowers migration frictions, allowing individuals to take their human capital to destinations where they earn higher wages. Output gains rise by 0.34 percentage points under this improved matching of individuals and jobs. Mobility reduces inequality across people, who have more equal access to job opportunities across space. But it also increases inequality across places, as migration concentrates human capital in high-wage destinations. Improved access to labor markets creates valuable opportunities for people, even as it increases the spatial concentration of economic activity.

The motives channel captures the feedback from migration opportunities to schooling, holding wage rates fixed. Greater access to job opportunities raises the return to schooling, and so schooling and migration reinforce each other: schooling lowers migration frictions, and lower migration frictions encourage schooling. Output gains rise by another 0.28 percentage points under this positive feedback loop. Access to other labor markets thus shapes both where individuals work and how much schooling they acquire.

The market channel captures the effects of equilibrium adjustments in wage rates. Increased human capital raises the supply of productive labor, generating congestion that lowers wage rates. These wage adjustments then induce further schooling and migration responses. Output gains fall by 1.21 percentage points. The market channel also partly reverses the distributional effects of mobility, increasing inequality across people and reducing inequality across places. Falling wage rates in destinations that attract human capital dampen both the gains to people who access these labor markets and the concentration of gains in those places.

Additional results describe the spatial distribution of gains and assess their sensitivity to estimated parameters. Appendix figure D1 shows that rural-born people and rural places benefit more than their urban counterparts. Appendix table D2 assesses sensitivity to (κ, θ), for which estimation relies more heavily on the structure of the

model. Estimation of (δ, ρ, λ) is instead tied directly to the quasi-experimental effects. This exercise is particularly important for κ, which is imprecisely estimated.[13] In all cases, school construction continues to raise output, reduce inequality across people, and increase inequality across places. For κ, lower values correspond to stronger congestion, which acts as a dampening force that meaningfully reduces output gains in equilibrium. Congestion has little effect on inequality across people, but it reduces the increase in inequality across places, again by dampening wage gains in districts that attract additional human capital. For θ, lower values correspond to stronger selection, and the magnitudes vary little for both output and inequality.

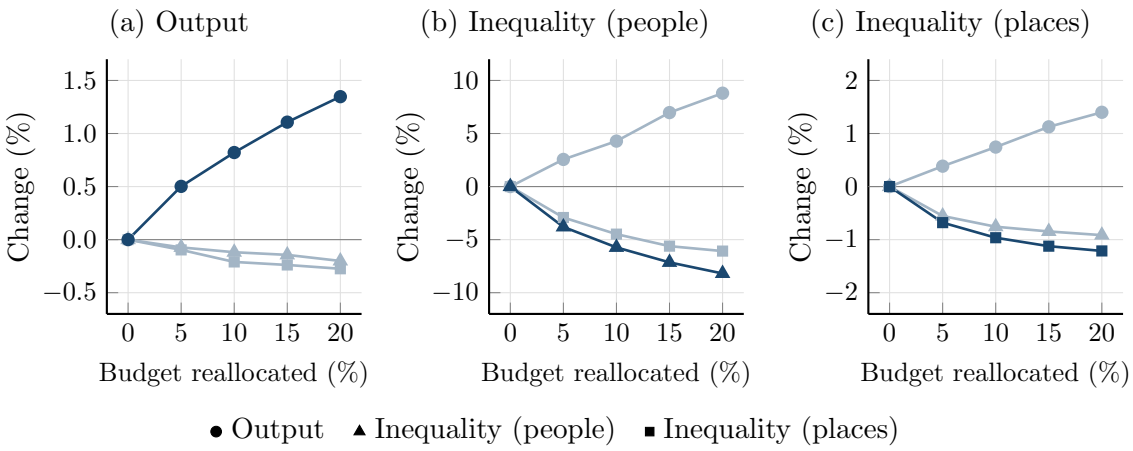
5.3 Investment

The spatial allocation of school construction shapes its aggregate and distributional effects, creating an equity-efficiency trade-off for the policymaker. I use the model to quantify this trade-off and examine its implications for the design of school construction policy. I consider three policy objectives: maximizing aggregate output, minimizing inequality across people, and minimizing inequality across places. Holding the national budget fixed, I allow 5%, 10%, 15%, and 20% of observed spending to be reallocated across districts, accounting for differences in school construction costs. I choose allocations jointly across districts because migration and equilibrium adjustments link the returns to construction across locations. To limit extrapolation, I keep construction per 1,000 children within the observed national range.

Output-maximizing reallocations raise output but also raise inequality. In figure 3, circles trace the effects of such reallocations across the outcomes shown in the three panels. Figure 3a shows that reallocating 10% of the budget raises output by 0.82% relative to observed construction, while reallocating 20% raises output by 1.35%. Figures 3b and 3c show the resulting distributional consequences. Reallocating 20% of the budget raises inequality across people by 8.80% and raises inequality across places by 1.40%. Larger budget reallocations yield larger output gains, while also exacerbating wage differences across people and places.

[13] I consider κ values from 0.6 to 1.0. Duflo (2004) reports wage elasticities with respect to formal-sector human capital of -0.34 to -0.40. Interpreting these values as $\kappa - 1$ gives $\kappa \in [0.60, 0.66]$. Bryan and Morten (2019) calibrate an agglomeration elasticity of 0.05 and an elasticity of substitution across regional goods of 8. The latter generates congestion through falling output prices. These values imply a net wage-rate elasticity of roughly $0.05 - 1/8 = -0.075$, yielding $\kappa = 0.925$.

Figure 3: Investment



I reallocate school construction to maximize output or minimize inequality across people or places, holding the total construction budget fixed. Each panel reports an outcome: output, inequality across people, or inequality across places. Markers distinguish the objective used to allocate construction: circles maximize output, triangles minimize inequality across people, and squares minimize inequality across places. Navy highlights the allocation that optimizes the outcome shown in each panel. The y-axis reports impacts on output and inequality in percentage changes relative to observed construction. The x-axis is the share of the national budget that I allow to be reallocated across districts. I calculate this share as half of the sum of absolute changes in district spending, divided by the observed budget. Output is aggregate wages. Inequality is the population-weighted Gini coefficient of district mean wages. Inequality across people compares mean wages of people born in different districts. Inequality across places compares mean wages of people living in different districts. The sample is male wage earners of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974.

By contrast, inequality-minimizing reallocations reduce inequality but also reduce output. Triangles trace allocations that minimize inequality across people, and squares trace those that minimize inequality across places. These two objectives have similar effects. In figure 3a, reallocating 20% of the budget reduces output by 0.20% under the people objective and 0.27% under the places objective. In figure 3b, these allocations reduce inequality across people by 8.19% and 6.08%. In figure 3c, they reduce inequality across places by 0.92% and 1.21%. Reallocation achieves a larger reduction in the specific inequality measure that it targets, but both objectives reduce inequality across people and places. Furthermore, with either objective, the policymaker reduces inequality at only modest cost to aggregate output.

Appendix figure D2 shows how these objectives change where schools are built, focusing on the 20% budget allowance. The observed allocation rule targeted districts with more unenrolled children. Maximizing output softens this targeting, shift-

ing construction toward districts with fewer unenrolled children but higher returns. Minimizing inequality instead reinforces this targeting, shifting construction toward districts with more unenrolled children despite lower returns. The choice of where to build schools therefore depends on how policymakers weigh aggregate and distributional gains. The observed allocation reflects a balance between these objectives.

6 Conclusion

Human capital is portable, and so the gains from school construction extend beyond the places where schools are built. Public investment in schools shapes individuals' investment in schooling. Schooling then facilitates migration, and migration opportunities encourage further schooling. I capture these forces with a spatial equilibrium model of schooling and migration, which I estimate with quasi-experimental variation from Indonesia's *Sekolah Dasar Inpres* program. I use the model to quantify the aggregate and distributional effects of school construction.

I find that new schools primarily benefit local students, but most of the resulting wage gains are earned elsewhere. Local outcomes capture only part of the gains from school construction. Furthermore, the *Inpres* program raised aggregate output and reduced inequality across people, while increasing inequality across places. Expanded opportunity for people came with greater disparities across places. At the same time, reallocating construction toward low-enrollment districts would have reduced inequality across both people and places, without large losses in aggregate output. Targeting disadvantaged districts can promote equity, while preserving most of the aggregate gains.

Several questions remain for future work. First, local investment generates gains beyond local borders. The political economy of these spillovers has important implications, particularly when investment is chosen and funded locally. Second, I assume that school construction lowers schooling costs by increasing physical access, but the effects of new schools may also depend on school quality and interactions with existing schools. Third, public school construction may prompt private sector responses that also affect aggregate outcomes. Such work informs policymakers' ongoing efforts to invest in education, which remains fundamental to economic development.

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Appendix

A Model

I present key derivations. For individual choice derivations, I fix an origin-cohort group (j, k) and suppress indices with notation $\tau_\ell = \tau_{jk\ell}(e)$, $\pi_\ell = \pi_{jk\ell}(e)$, $\bar{r} = \bar{r}_{jk}(e)$, $\Omega = \Omega_{jk}(e)$, $m = m_{jk}(e)$, and $M = M_{jk}(e)$.

Selection

I derive the selection expression in equation 16 and its log moments. Productivity draws follow a Fréchet distribution with shape θ and mean one.

$$F(\varepsilon) = \exp\{-(\varepsilon/\sigma)^{-\theta}\}, \quad \sigma = \frac{1}{\Gamma(1 - 1/\theta)}$$

for scale σ and gamma function $\Gamma(a) = \int_0^\infty t^{a-1} e^{-t}\, dt$. Choosing ℓ requires

$$\frac{r_{k\ell}\varepsilon_{i\ell}}{\tau_\ell} \geq \frac{r_{k\ell'}\varepsilon_{i\ell'}}{\tau'_\ell} \quad \forall\, \ell' \neq \ell.$$

Independence across destinations gives

$$\begin{aligned}
\Pr(\varepsilon_{i\ell} \leq x, \ell_i = \ell) &= \int_0^x \prod_{\ell' \neq \ell} F\left(z \frac{r_{k\ell}/\tau_\ell}{r_{k\ell'}/\tau_{\ell'}}\right) dF(z) \\
&= \int_0^x \theta \sigma^\theta z^{-\theta-1} \exp\left\{-\Omega \left(\frac{z r_{k\ell}}{\sigma \tau_\ell}\right)^{-\theta}\right\} dz \\
&= \frac{1}{\Omega}\left(\frac{r_{k\ell}}{\tau_\ell}\right)^\theta \exp\left\{-\Omega \left(\frac{x r_{k\ell}}{\sigma \tau_\ell}\right)^{-\theta}\right\}
\end{aligned}$$

for $x > 0$ and access Ω in equation 5. The first equality follows from the condition for choosing ℓ. The second substitutes the Fréchet distribution and the definition of access, and the third integrates. Taking $x \to \infty$ gives the choice probabilities in equation 6. Dividing the joint probability by π_ℓ gives a conditional Fréchet distribution with shape θ and scale $\sigma \pi_\ell^{-1/\theta}$. The conditional expectation is

$$\mathbb{E}\left[\varepsilon_{i\ell} \,|\, \ell_i = \ell\right] = \sigma \pi_\ell^{-1/\theta} \Gamma\left(1 - \frac{1}{\theta}\right) = \pi_\ell^{-1/\theta}, \tag{29}$$

where the last equality uses the mean-one normalization $\sigma\Gamma(1-\frac{1}{\theta})=1$.

For log expressions, define $G=\theta\log(\frac{\varepsilon_{i\ell}}{\sigma})+\log\pi_\ell$. The conditional distribution implies

$$\Pr(G\le g\,|\,\ell_i=\ell)=\Pr\Big(\varepsilon_{i\ell}\le\sigma\pi_\ell^{-1/\theta}e^{g/\theta}\,|\,\ell_i=\ell\Big)=\exp\{-e^{-g}\}.$$

Thus, conditional on choosing ℓ, G is Gumbel-distributed with mean γ_E and variance $\frac{\pi^2}{6}$, where γ_E is Euler's constant. I rewrite the definition as $\log\varepsilon_{i\ell}=\log\sigma-\frac{1}{\theta}\log\pi_\ell+\frac{G}{\theta}$ and take the conditional mean and variance.

$$\mathbb{E}[\log\varepsilon_{i\ell}\,|\,\ell_i=\ell]=c_\theta-\frac{1}{\theta}\log\pi_\ell,\quad \mathrm{Var}(\log\varepsilon_{i\ell}\,|\,\ell_i=\ell)=\frac{\pi^2}{6\theta^2},\tag{30}$$

where $c_\theta=\log\sigma+\frac{\gamma_E}{\theta}=\frac{\gamma_E}{\theta}-\log\Gamma(1-\frac{1}{\theta})$.

Migration

I derive the probability of leaving home and expected utility in equations 7 and 8. Applying $\tau_j=1$ to the choice probabilities in equation 6 gives home share $\pi_j=\frac{1}{\Omega}r_{kj}^\theta$. Equation 7 then follows from $m=1-\pi_j$. For equation 8, expected utility per unit of human capital is

$$\bar r=\sum_\ell\frac{\pi_\ell r_{k\ell}}{\tau_\ell}\mathbb{E}[\varepsilon_{i\ell}\,|\,\ell_i=\ell]=\sum_\ell\frac{r_{k\ell}}{\tau_\ell}\pi_\ell^{1-1/\theta}=\Omega^{1/\theta}.$$

The first equality applies the law of iterated expectations, the second substitutes equation 29, and the third substitutes the choice probabilities.

Education

I derive the schooling condition in equation 10. Schooling raises expected utility by reducing migration frictions. Differentiating $\bar r=\Omega^{1/\theta}$ at fixed wage rates gives

$$\frac{d\log\bar r}{de}=\sum_\ell\pi_\ell\left(-\frac{\partial\log\tau_\ell}{\partial e}\right)=\frac{\lambda}{\theta}m.$$

The first equality weights reductions in frictions by choice probabilities. The second applies equation 4, by which the reduction is $\frac{\lambda}{\theta}$ away from home and zero at home. The odds of leaving home grow at rate λ, and so $\frac{dm}{de}=\lambda m(1-m)$.

Schooling raises expected utility tomorrow but incurs costs today. Differentiating the objective function of equation 9 and applying $h_i'(e) = \rho h_i(e)$ and $\frac{d\log \bar{r}}{de} = \frac{\lambda m}{\theta}$ as derived above, I obtain

$$c_i h_i(e_i) = \bar{r}(e_i)\left[1 + \frac{\lambda}{\rho\theta} m(e_i)\right] \tag{31}$$

at an interior optimum. Taking logs and applying equations 7 and 8 gives the schooling condition in equation 10 and the away term in equation 11. At fixed wage rates, optimal schooling is unique if $\rho > \frac{d\log \Lambda}{de}$ and $\rho > \frac{\lambda}{\theta}$ for all $e \geq 0$.

Labor demand

Firms take the local productivity component $A_\ell H_\ell^{\kappa-1}$ as given. The marginal product of cohort k's human capital is therefore

$$r_{k\ell} = A_\ell H_\ell^{\kappa-1} \frac{\alpha_{k\ell} H_\ell}{H_{k\ell}} = \alpha_{k\ell} A_\ell \frac{H_\ell^\kappa}{H_{k\ell}},$$

which gives equation 14. Substituting into the wage index gives

$$R_\ell = \prod_k \left(\frac{r_{k\ell}}{\alpha_{k\ell}}\right)^{\alpha_{k\ell}} = A_\ell H_\ell^{\kappa-1}.$$

Combining the two expressions yields

$$r_{k\ell} H_{k\ell} = \alpha_{k\ell} R_\ell H_\ell.$$

For $\kappa \neq 1$, solving $R_\ell = A_\ell H_\ell^{\kappa-1}$ for aggregate demand and substituting it into this expression gives equation 18. For $\kappa = 1$, the wage index satisfies $R_\ell = A_\ell$, while relative cohort demands remain inversely related to relative wage rates.

Labor supply

By equation 29, individual i supplies expected effective labor $\pi_{i\ell}\,\mathbb{E}[h_i \varepsilon_{i\ell} \mid \ell_i = \ell] = h_i \pi_{i\ell}^{1-1/\theta}$ to location ℓ. Summing over individuals $i \in \mathcal{I}_k$ with their choice probabilities gives equation 17. The same formula gives expected wages $\bar{w}_i = \sum_\ell r_{k\ell} h_i \pi_{i\ell}^{1-1/\theta}$ in equation 21. The wage bill is $W_\ell = \sum_k r_{k\ell} H_{k\ell}$, which equals Y_ℓ by equation 15.

Motives

I derive the motives multiplier in equation 23. For an individual with positive schooling, I substitute the cost shifter of equation 20 into the schooling condition of equation 10, and differentiate at fixed wage rates to obtain

$$d\log h_i = \rho\, de_i = \delta T_k\, dS_j + \frac{d\log\Lambda}{de}\, de_i.$$

Collecting the de_i terms and dividing by dS_j, holding T_k fixed, yields

$$\frac{\partial e_i}{\partial S_j} = \frac{\delta T_k}{\rho - d\log\Lambda/de} = \frac{\delta}{\rho} T_k M_i.$$

For treated cohorts, $T_k = 1$ gives the schooling response $\frac{\delta}{\rho} M_i$. To evaluate the $\log\Lambda$ derivative, I differentiate the away term of equation 11 and substitute $\frac{dm}{de} = \lambda m(1-m)$ as derived above.

$$\frac{d\log\Lambda}{de} = \left[\frac{1}{\theta(1-m)} + \frac{\lambda}{\rho\theta + \lambda m}\right]\frac{dm}{de} = \frac{\lambda}{\theta} m + \frac{\lambda^2 m(1-m)}{\rho\theta + \lambda m} \geq 0$$

The second-order condition $\rho > \frac{d\log\Lambda}{de}$ ensures that the multiplier has a positive denominator. Thus, $M_i \geq 1$ with equality when $\lambda = 0$ or when no one leaves home.

Mobility

I derive the mobility return in equation 23. By equations 3 and 29, expected wages among individuals choosing ℓ are

$$\bar{w}_{i\ell} = \mathbb{E}\big[w_{i\ell} \,|\, \ell_i = \ell\big] = r_{k\ell} h_i \pi_\ell^{-1/\theta} = \bar{r} h_i \tau_\ell.$$

Differentiating the choice probabilities of equation 6 at fixed wage rates,

$$\frac{d\log\pi_\ell}{de} = -\theta\frac{d\log\tau_\ell}{de} - \frac{d\log\Omega}{de} = \lambda\big(\mathbb{1}\{\ell \neq j\} - m\big).$$

The second equality uses the frictions in equation 4 and $\frac{d\log\Omega}{de} = \theta\frac{d\log\bar{r}}{de} = \lambda m$. Differentiating the expected wage contribution from destination ℓ,

$$\frac{d\log(\pi_\ell \bar{w}_{i\ell})}{de} = \rho + \left(1 - \frac{1}{\theta}\right)\frac{d\log\pi_\ell}{de} = \rho + \lambda\left(1 - \frac{1}{\theta}\right)(\mathbb{1}\{\ell \neq j\} - m).$$

The first equality differentiates $\pi_\ell \bar{w}_{i\ell} = r_{k\ell} h_i \pi_\ell^{1-1/\theta}$, and the second substitutes the migration response. Since $\bar{w}_i = \sum_\ell \pi_\ell \bar{w}_{i\ell}$, its log derivative averages these derivatives with wage shares $s_\ell = \frac{\pi_\ell \bar{w}_{i\ell}}{\bar{w}_i}$.

$$\frac{d\log\bar{w}_i}{de} = \sum_\ell s_\ell \frac{d\log(\pi_\ell \bar{w}_{i\ell})}{de} = \rho + \lambda\left(1 - \frac{1}{\theta}\right)(\tilde{m} - m),$$

where $\tilde{m} = \sum_{\ell\neq j} s_\ell$ and $\sum_\ell s_\ell = 1$. The term beyond the direct human-capital return ρ is the mobility return μ_i in equation 23. To establish its sign, use $\bar{w}_{i\ell} = \bar{r} h_i \tau_\ell$ and $\tau_j = 1$ to write

$$\tilde{m} = \frac{\sum_{\ell\neq j}\pi_\ell\tau_\ell}{1 - m + \sum_{\ell\neq j}\pi_\ell\tau_\ell}, \quad \tilde{m} - m = \frac{(1-m)\left(\sum_{\ell\neq j}\pi_\ell\tau_\ell - m\right)}{1 - m + \sum_{\ell\neq j}\pi_\ell\tau_\ell}.$$

For $\lambda > 0$ and a positive home share, $\mu_i > 0$ if and only if $\sum_{\ell\neq j}\pi_\ell\tau_\ell > m$. This condition captures that migration frictions are net costs. Finally, at fixed wage rates and an interior schooling solution, multiplying wage return $\rho+\mu_i$ by schooling response $\frac{\delta}{\rho}M_i$ gives the decomposition in equation 22 by the chain rule.

B Data

I describe the data and data construction. Table B1 lists data sources.

B.1 Treatment data

Data on the *Inpres* program and district demographics come from Duflo (2001), which draws on Ministry of National Development Planning (*Bappenas*) reports from 1973 to 1978 and population census reports from 1971. For each district, the data record the planned number of *Inpres* schools by year from 1973 to 1978, *Inpres* water and sanitation spending from 1973 to 1978, total and child populations in 1971, and child enrollment rates in 1971. The data also report district-level construction spending in 1974, which I divide by school counts to obtain costs per school. From the original dataset, I drop duplicate and missing observations to obtain measures for 282 unique districts. Districts include rural regencies (*kabupaten*) and urban municipalities (*kota*). Figure B1 plots school construction against unenrollment and population density across districts. I verify the data and make adjustments as follows.

I consult the original *Bappenas* reports to verify the school construction data, and I correct a modest number of data entry errors and inconsistencies: 1 for 1973, 4 for 1974, 12 for 1975, 5 for 1976, 3 for 1977, and 2 for 1978. Among these changes, the corrected values are on average 85% larger or smaller than their original values. In the corrected data, construction in 1976 and 1978 matches that in 1975 and 1977, respectively, as stipulated by the presidential instructions.[14] I take water and sanitation spending to be as constructed by Duflo (2001), noting that this spending seems to be proportional to school construction in the *Bappenas* reports, and I fill 6 missing observations to avoid dropping these districts.[15] For total and child populations, I again consult the original census reports and make a small number of corrections: 29 for total population and 16 for child population. The average magnitudes of these changes are 65% and 38% of the original values. Child populations are defined to

[14] In reference to *Inpres* No. 6/1975, *Inpres* No. 3/1976 states: "*Pembagian jumlah gedung Sekolah Dasar [...] adalah sama dengan jumlah gedung Sekolah Dasar tahap pertama yang dibangun [...] berdasarkan Instruksi Presiden Nomor 6 Tahun 1975.*" *Inpres* No. 6/1978 contains analogous text in reference to *Inpres* No. 3/1977.

[15] I impute 1 missing value for Southeast Aceh with the average value among other rural districts in the province of Aceh. I fill 5 missing values for the Special Capital Region of Jakarta with zero.

include individuals of age 5 to 14. With *Inpres* school construction from 1973 to 1978 and child populations for 1971, I recompute the main treatment variable as the total number of *Inpres* schools constructed per 1,000 children.

I construct child enrollment rates with census microdata from 1971, drawing on a 0.54% sample from IPUMS International (Ruggles et al. 2024). These microdata record whether respondents are currently in school, and I compute enrollment rates among individuals of age 5 to 14 to match those captured by the child population data. I do so for the 275 districts I observe in the IPUMS data. Redistricting between 1971 and 1973 explains the difference from the 282 districts of the school construction data. I fill the 7 missing observations for (offshoot) child districts by applying the enrollment values of the corresponding (original) parent districts. I obtain rather different measures of child enrollment, and the average magnitude of the changes is 221% of the original values.

Table B2 summarizes the data adjustments relative to the original data from Duflo (2001). The baseline data are balanced with a slightly smaller number of districts. *Inpres* school construction and populations remain similar on average. I obtain higher child enrollment rates, as the Duflo (2001) data seem to define child enrollment rates based on school attendance for individuals of age 5 and above, rather than individuals of age 5 to 14. I obtain similar magnitudes when I construct enrollment rates based on the former sample of individuals. At the same time, I note that child enrollment enters the analysis only as a control variable. I also note that the regressions in Duflo (2001) may not rely on this particular measure, as the data also record the number of children attending school.

I construct population density from total populations and land areas. I use district boundary shapefiles from the Database of Global Administrative Areas (GADM) to compute land area by district, as well as district centroids. Because of redistricting over time, some district polygons must be merged to obtain the district boundaries of the treatment data. Merged land areas sum over individual land areas, and merged centroids are area-weighted averages of individual centroids.

B.2 Outcome data

Data on individual outcomes come from the National Socioeconomic Survey (*Susenas*) for 2011, 2012, 2013, and 2014. I observe demographics, schooling, and wages. Demographics include gender, age, district of birth, and district of residence. Following Duflo (2001), I restrict attention to male individuals of age 2 to 24 in 1974, observed at ages 39 to 64 across the survey years. I use district of birth to link individuals to *Inpres* school construction, assuming that individuals pursue primary schooling where they are born. I use district of residence to construct migration outcomes, with migrants defined as those whose districts of birth and residence do not coincide.

I encounter the typical challenge of redistricting over time, as decentralization has prompted many cases of larger districts that split into smaller, independent districts. I merge districts as coded in the *Susenas* data to match those coded in the *Inpres* school construction data. I do so with a district proliferation crosswalk from the World Bank, combined with manual inspection. I obtain a consistent set of boundaries over time, covering 26 provinces and 282 districts.

For schooling, I observe the highest level attended and either completion or the number of years attended at that level. I pool vocational, religious, and traditional schooling for middle school and high school. At the tertiary level, I distinguish between community colleges that offer vocational associate's degrees (*Diploma*) and universities that offer traditional bachelor's degrees (*Sarjana*). I compute the years of schooling completed. Primary school is six years, middle school is three, high school is three, community college is two, and university is four. For those who attend but do not complete a given level of schooling, I subtract one from the number of years attended at that level.[16] I code zero years of schooling for those who report never having attended school in a preliminary question.

Tertiary schooling is a challenge. Three-year vocational programs exist (*D3*), but the *Susenas* data pool these programs with associate's degrees (*Sarjana Muda* in older parlance) that may not be three-year programs. The data also pool master's and doctoral programs (*S2* and *S3*), which typically require different periods of study.

[16] For example, for someone who reports attending three years of middle school but not completing middle school, I record eight years of schooling completed rather than nine years attended.

Attendance without completion is more common at this level, with recall bias serving to worsen measurement error. Furthermore, the *Inpres* program built primary schools, which in any case may have limited impacts on tertiary schooling outcomes. The analysis takes twelve years of schooling as the maximum, sidestepping the potential measurement error from tertiary schooling.

For employment, I observe employment status, type, income, and hours. I code individuals as employed if they report working as their primary activity in the past week. Individuals also report whether they work as own-account workers, employers with temporary or permanent employees, temporary or permanent employees, or unpaid family workers. I categorize employees as wage employed and others as non-wage employed. I observe typical monthly income from an individual's main job, inclusive of both money and goods, and I adjust for inflation to obtain income in year-2011 rupiah. I also observe the number of hours worked in the past week at an individual's main job, which I combine with monthly income to compute log hourly income.

A data limitation is that the income data are incomplete for non-wage employed individuals. Table B3 shows this incompleteness by tabulating the income data by employment type. For the wage employed, I record nonzero incomes for 90% of individuals. These data are nearly complete. But for the non-wage employed, I record nonzero incomes for only 59% of individuals. I observe that reported incomes are lower among the non-wage employed, for whom income is inherently more variable and perhaps prone to reporting bias. My analysis focuses on income for the wage employed, and I refer to wages in place of income.

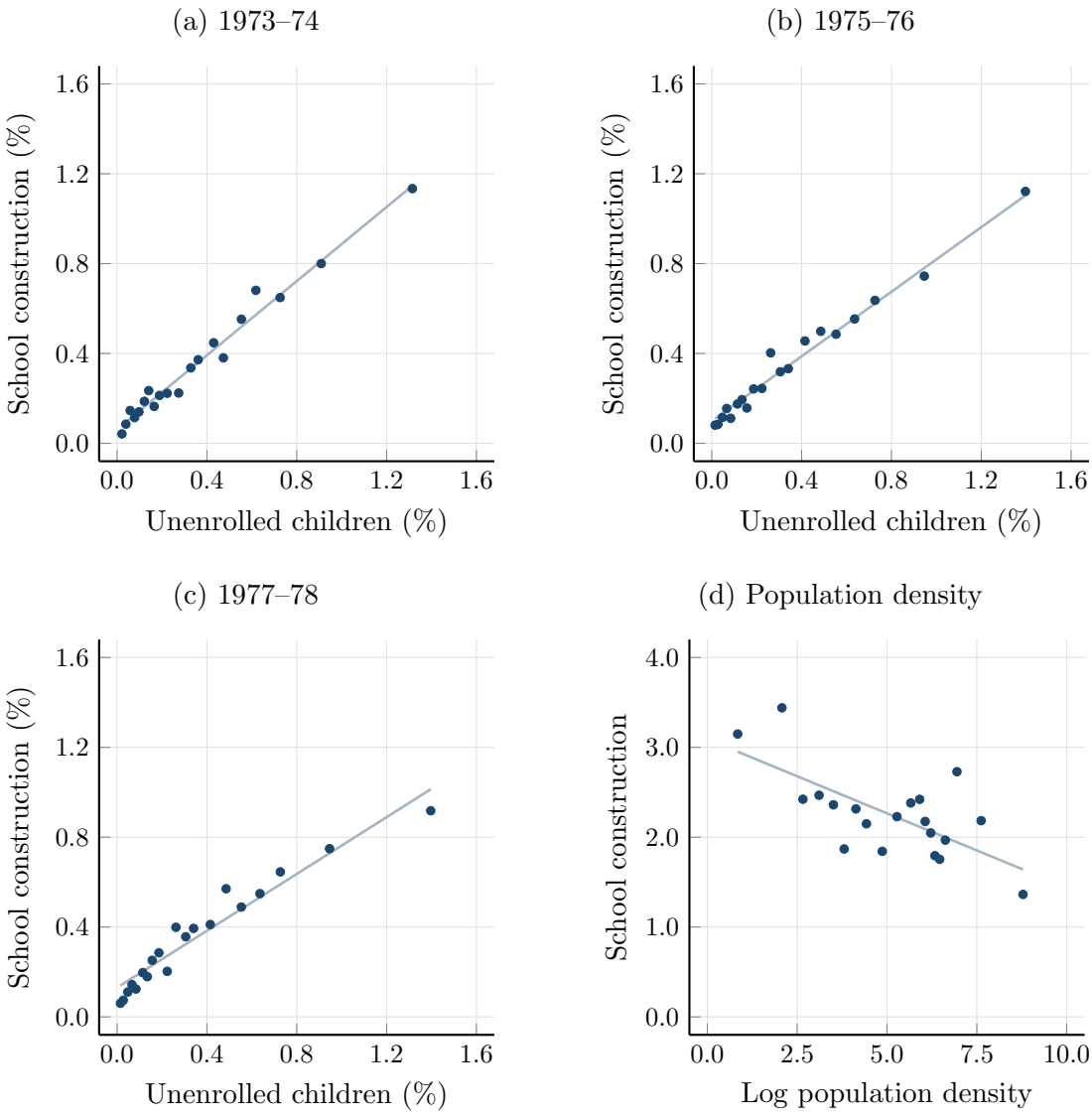
Data on historical labor market outcomes come from the National Labor Force Survey (*Sakernas*) for 1986, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1996, and 1997. I retain male individuals of age 20 to 60 in each survey year, and I use these repeated cross-sections to construct a district panel of wages and labor supply, harmonized to 1973 boundaries. I use survey weights throughout. For wages, I restrict attention to unexposed cohorts born by 1962 and compute mean log hourly wages among wage earners. Hourly wages use wages and hours at the main job. I cap wages at each year's 99.5th percentile among those with positive wages, and I report wage levels in 2011 prices using the national consumer price index.

For labor supply, I measure the total years of schooling embodied in wage employment. I map completed schooling levels to years as I do for the *Susenas* data, capping at twelve years, and I assign three years to individuals with incomplete primary schooling. For each district, I compute the mean years of schooling in wage employment per man of age 20 to 60, assigning zero to those who are not wage employed. I then multiply by census estimates of total district populations from Badan Pusat Statistik (2000). I obtain year-specific district populations by interpolating linearly between the 1990 and 2000 census measures and extending the same linear trend back to 1986. I further multiply by each district's share of men of age 20 to 60 in the 1971 census, holding this share fixed across survey years.

Table B1: Data sources

Period	Format	Source	Description
Treatment data			
1973–1978	Reports	Ministry of National Development Planning (*Bappenas*), Duflo (2001)	*Inpres* school construction and water and sanitation spending
1971	Reports	Population Census (*SP*), Duflo (2001)	Total and child populations
1971	Microdata	Population Census (*SP*), Ruggles et al. (2024)	Child enrollment
2011	Shapefiles	Database of Global Administrative Areas (GADM)	Administrative boundaries (level 2)
1993–2014	Crosswalk	World Bank	Redistricting over time
Outcome data			
2011–2014	Microdata	National Socioeconomic Survey (*Susenas*)	Schooling, wages, employment, and migration
1986–1997	Microdata	National Labor Force Survey (*Sakernas*)	District wages and labor supply
1990, 2000	Reports	Population Census (*SP*), Badan Pusat Statistik (2000)	District populations

Figure B1: Program allocation



Each panel is a binned scatter plot across the 282 program districts. In panels (a) to (c), the y-axis is a district's share of total school construction in each period. The x-axis for 1973–74 is a district's share of all unenrolled children of primary school age in 1971. For later years, it counts unenrolled children in excess of 15% of each district's child population. Panel (d) plots school construction per 1,000 children from 1973 to 1978 against log population density in 1971.

Table B2: Treatment data adjustments

	Baseline			Duflo (2001)		
	Mean	SD	N	Mean	SD	N
Inpres schools per 1,000 children	2.26	1.09	282	2.34	1.26	287
Inpres school construction	220	175	282	222	173	293
1973 construction	21.3	20.0	282	21.5	20.1	286
1974 construction	21.3	18.5	282	21.5	18.5	286
1975 construction	35.5	29.9	282	35.5	29.8	286
1976 construction	35.5	29.9	282	35.5	29.8	286
1977 construction	53.1	44.6	282	53.1	44.2	285
1978 construction	53.1	44.6	282	53.2	44.3	285
Population (1000s)	423	362	282	420	366	287
Child population (1000s)	115	98.0	282	115	98.0	287
Child enrollment rates	0.51	0.14	282	0.18	0.10	287

Each observation is a district. Columns show means, standard deviations, and the number of observations in the baseline data and the Duflo (2001) data. Rows summarize the number of *Inpres* schools constructed from 1973 to 1978 and population and enrollment in 1971. Children are those of age 5 to 14.

Table B3: Income data

	Log income	Reported income	N
Wage employed	9.12	0.90	181,393
Non-wage employed	8.86	0.59	257,516

Each observation is an employed male individual of age 2 to 24 in 1974, observed at ages 39 to 64 across the survey years. Columns show mean log income conditional on reporting nonzero income, the proportion of individuals who report nonzero income, and the number of observations. Income is log hourly net income in year-2011 Indonesian rupiah from an individual's main job. Rows summarize these statistics for the wage and non-wage employed.

C Estimation

I describe the estimation procedure and report additional results.

C.1 Schooling, wages, and migration

The implied parameters in table 3 jointly match the schooling, wage, and migration coefficients of columns (1), (2), and (3).[17] I estimate parameters $\psi = (\delta, \rho, \lambda)'$ as follows, conditioning on the κ estimate of table 4 and the θ estimate of table 5. First, for a candidate ψ, I use the model to calculate *Inpres* treatment effects as the difference between outcomes y_i^1 under observed school construction and outcomes y_i^0 under no school construction.

$$\Delta y_i(\psi) = y_i^1(\psi) - y_i^0(\psi)$$

I compute these objects by solving the model as described in appendix D.

Second, I relate the model-implied effects to the target coefficients. The assumption of parallel trends implies that, absent the program, cohort differences would be unrelated to construction intensity after accounting for fixed effects and controls. Thus, by equation 26,

$$\begin{aligned} y_i^1 &= \alpha_j^1 + \alpha_k^1 + \alpha_t^1 + \beta S_j T_k + T_k C_j' \gamma^1 + \varepsilon_i^1, \\ y_i^0 &= \alpha_j^0 + \alpha_k^0 + \alpha_t^0 + T_k C_j' \gamma^0 + \varepsilon_i^0, \end{aligned}$$

where ε_i^0 is uncorrelated with $S_j T_k$ conditional on fixed effects and controls. Differencing yields

$$\Delta y_i = \Delta\alpha_j + \Delta\alpha_k + \Delta\alpha_t + \beta S_j T_k + T_k C_j' \Delta\gamma + \Delta\varepsilon_i.$$

For each outcome, this specification identifies the same coefficient β that I estimate in table 3. For each candidate ψ, I compute model-implied effects $\Delta y_i(\psi)$ and estimate this specification with the original regression sample, fixed effects, and controls. I choose parameters that match the target coefficients.

[17] It is equivalent to match the schooling, wage IV, and migration IV coefficients of columns (1), (8), and (10) because I match the coefficients exactly.

C.2 Equilibrium

I derive the equilibrium regression and describe the human capital measures. Mean log wages among unexposed cohorts satisfy

$$\log w_{U\ell t} = (\kappa - 1) \log H_{\ell t} - \log \frac{H_{U\ell t}}{H_{\ell t}} + \log A_{U\ell t},$$

where $H_{\ell t}$ is aggregate human capital, $H_{U\ell t}$ is human capital among unexposed cohorts, and $A_{U\ell t}$ combines productivity, production weights, and sample composition. The expression combines equations 3 and 14 and averages over unexposed wage earners in district ℓ and survey year t, such that $\log A_{U\ell t} = \log A_{\ell t} + \log \alpha_{U\ell} + \mathbb{E}_{U\ell t}[\log h_i(e_i) + \log \varepsilon_{i\ell}]$. On the right-hand side, the first term captures congestion and agglomeration. The second term captures relative supply effects, by which growth in exposed relative to unexposed labor supply raises wages for unexposed cohorts. I rearrange and decompose the productivity term as $\log A_{U\ell t} = \alpha_\ell + \alpha_t + C'_\ell \gamma_t + \varepsilon_{\ell t}$ to obtain estimating equation 27. Identification requires the instrument to be uncorrelated with $\varepsilon_{\ell t}$ conditional on fixed effects and controls.

I construct the human capital terms from the data. I measure human capital $H_{k\ell t}$ for exposed and unexposed cohorts $k \in \{E, U\}$ with the years of schooling embodied in wage employment, as described in appendix B. To aggregate across cohorts, I recover production weights from the cohort shares of the wage bill. Equation 14 implies

$$\alpha_{E\ell} = \frac{W_{E\ell t}}{W_{E\ell t} + W_{U\ell t}},$$

where $W_{k\ell t}$ is the observed wage bill for cohort k. I measure these cohort shares before 1992, pooling wages across survey years, and I hold them fixed. Aggregate human capital then follows from equation 13.

$$\log H_{\ell t} = \alpha_{E\ell} \log H_{E\ell t} + (1 - \alpha_{E\ell}) \log H_{U\ell t}$$

C.3 Selection

I derive the selection regression from the model's wage equation. For individual i born in district j and cohort k, with schooling e_i and observed in destination ℓ in

survey year t, equations 1 and 3 imply observed wages $w_i = w_{i\ell} = r_{k\ell t} a_i s_{jk} \exp(\rho e_i)\varepsilon_{i\ell}$. Taking logs,

$$\log w_i = \rho e_i + \log s_{jk} + \log r_{k\ell t} + \log a_i + \log \varepsilon_{i\ell}.$$

I define two deviations from conditional means.

$$\nu_i^{\varepsilon} = \log \varepsilon_{i\ell} - \mathbb{E}_{jk\ell t}\big[\log \varepsilon_{i\ell} \,|\, e_i\big], \quad \nu_i^{a} = \log a_i - \mathbb{E}_{jk}\big[\log a_i \,|\, e_i\big],$$

where $\mathbb{E}_{jk\ell t}[\cdot \,|\, e_i]$ and $\mathbb{E}_{jk}[\cdot \,|\, e_i]$ denote conditional means and ℓ denotes the chosen destination. By equation 30, the Fréchet distribution implies

$$\mathbb{E}_{jk\ell t}\big[\log \varepsilon_{i\ell} \,|\, e_i\big] = c_\theta - \frac{1}{\theta} \log \pi_{jk\ell t}(e_i),$$

where constant c_θ is as defined in appendix A. Conditional on origin, cohort, and schooling, location choices are independent of ability: ability scales wages equally across destinations, and destination productivity draws are independent of ability. Conditional mean log ability is therefore common across destinations within these groups. Substitution yields

$$\log w_i = \underbrace{\rho e_i + \log s_{jk} + c_\theta + \mathbb{E}_{jk}\big[\log a_i \,|\, e_i\big]}_{\alpha_{jk}(e_i)} + \underbrace{\log r_{k\ell t}}_{\alpha_{k\ell t}} - \frac{1}{\theta} \log \pi_{jk\ell t}(e_i) + \nu_i,$$

where $\nu_i = \nu_i^{\varepsilon} + \nu_i^{a}$ has conditional mean zero. Replacing choice probabilities with empirical shares gives estimating equation 28.

I construct empirical location shares within groups defined by birth district, cohort, and schooling. Individuals of age 2 to 6 in 1974 are exposed, while those of age 12 to 17 are unexposed. I pool years of schooling into bands of 0 to 6, 7 to 9, and 10 to 12, with schooling capped at 12 years, and I pool observations across survey years from 2011 to 2014. Pooling reduces noise by increasing the number of observations per group. Within each group, location shares are the proportions of individuals who choose each destination. Pooling applies only to the construction of location shares, and the selection regression retains the full set of granular fixed effects. Table C4 assesses robustness to each pooling choice. I also evaluate sensitivity to dropping sparse destination-group cells by excluding those with fewer than five observations, recalculating location shares, and reestimating the selection regression.

Table C1: Controls

	Estimate	SE	N
Duflo (2001)			
Years of schooling	0.081**	(0.036)	264,307
Years of schooling	0.098**	(0.040)	98,413
Log wages	0.026***	(0.010)	98,413
With population decile			
Years of schooling	0.122***	(0.037)	264,307
Years of schooling	0.124***	(0.041)	98,413
Log wages	0.024**	(0.010)	98,413
With island group			
Years of schooling	0.095**	(0.041)	264,307
Years of schooling	0.113***	(0.043)	98,413
Log wages	0.030***	(0.009)	98,413
With both			
Years of schooling	0.138***	(0.039)	264,307
Years of schooling	0.148***	(0.042)	98,413
Log wages	0.031***	(0.009)	98,413

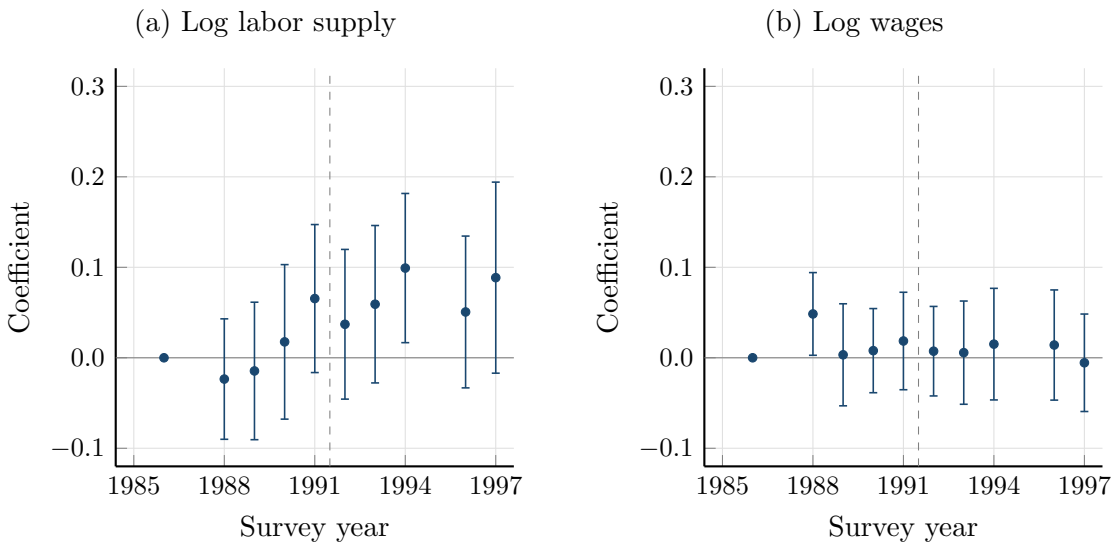
Each row is a regression, and each observation is a male individual of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974. Schooling is years completed in the full sample and the wage sample, and wages are log hourly wages in the wage sample. In the first block, I include fixed effects for birth district, birth year, and survey year, as well as controls for child population, child enrollment, and *Inpres* water and sanitation spending, as in Duflo (2001). Subsequent blocks add controls for population decile and island group. I interact all controls with exposure T_k, and I cluster standard errors by birth district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C2: Treatment intensity

		Intensity quartile				
	Linear	Q1	Q2	Q3	Q4	p-value
Years of schooling	0.148***	0.099	0.245**	0.158*	0.142**	0.07
	(0.042)	(0.158)	(0.117)	(0.088)	(0.056)	
Log wages	0.031***	−0.005	0.021	0.022	0.021	0.33
	(0.009)	(0.039)	(0.028)	(0.022)	(0.014)	
Migration	1.23**	3.59*	2.39	2.76**	1.70**	0.18
	(0.54)	(2.15)	(1.55)	(1.16)	(0.80)	
Weight in linear coefficient	1.00	−0.39	−0.16	0.36	1.19	
Schools per 1,000 children		1.21	1.76	2.37	3.48	

Each row is an outcome, and each observation is a male wage earner of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974. Schooling is years completed, wages are log hourly wages, and migration is residence outside of the birth district in percentage points. The linear column reports the treatment effects of table 3a. The Q1 to Q4 columns allow the effect to differ by quartile of construction intensity across districts, estimated jointly in one regression per outcome. The linear coefficient equals the sum of the quartile coefficients times the weights in the second-to-last row, where each weight is the quartile's share of $\sum_i \tilde{z}_i z_i$ for $z_i = S_j T_k$ and its residual $\tilde{z}_i$ after projecting on the fixed effects and controls. The p-value tests that the four quartile coefficients are equal. I include fixed effects for birth district, birth year, and survey year, as well as controls for child population and enrollment, *Inpres* water and sanitation spending, population decile, and island group. I interact all controls with exposure T_k, and I cluster standard errors by birth district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure C1: Timing of equilibrium effects



Each panel is a regression, and each observation is a district-year cell of male individuals of age 20 to 60. Labor supply is log aggregate human capital, constructed from total years of schooling in wage employment by cohort. Wages are mean log hourly wages of unexposed cohorts born by 1962, adjusted for relative labor supply. The y-axis reports coefficients on construction intensity S_ℓ interacted with indicators for each survey year, relative to 1986, with 95% confidence intervals. The x-axis is survey year. Exposed cohorts reach age 20 from 1988 to 1992, and the dashed line marks entry E_t from 1992. I include fixed effects for district and survey year, as well as controls for child population and enrollment, *Inpres* water and sanitation spending, population decile, and island group. I interact all controls with survey year, and I cluster standard errors by district.

Table C3: Robustness of equilibrium effects

	Log wages				
		Cohort human capital		Production weights	
	Baseline	Wage weights	Exponential weights	Annual shares	Population scaling
	(1)	(2)	(3)	(4)	(5)
Log labor supply	−0.153	−0.085	−0.128	−0.095	−0.157
	(0.255)	(0.236)	(0.233)	(0.263)	(0.254)
AR 95% set	[−1.19, 0.39]	[−0.83, 0.48]	[−0.91, 0.34]	[−1.25, 0.56]	[−1.19, 0.38]
First-stage F	6.4	7.2	7.9	5.7	6.5
Observations	2,596	2,596	2,596	2,596	2,596
Control mean	8.59	8.60	8.59	8.58	8.59

Each column is a regression, and each observation is a district-year cell of male individuals of age 20 to 60. Labor supply is log aggregate human capital. Wages are mean log hourly wages of unexposed cohorts born by 1962, adjusted for relative labor supply. Column (1) reports the IV estimate of table 4b. To construct cohort-specific human capital, column (2) weights wage employment by the estimated relative wage of each education group in each year, rather than weighting by years of schooling. I estimate relative wages from a regression of log hourly wages on fixed effects for district by year and education group by year. Column (3) instead weights wage employment by exponential weights $\exp(\rho e)$. To construct aggregate human capital, column (4) computes local production weights by averaging over annual cohort wage shares, rather than computing shares from pooled cohort wages. Column (5) instead scales wages by district population before computing pooled shares. I include fixed effects for district and survey year, as well as controls for child population and enrollment, *Inpres* water and sanitation spending, population decile, and island group. I interact all controls with survey year, and I cluster standard errors by district. Each column instruments for labor supply with $S_\ell E_t$ and reports Anderson–Rubin confidence sets. Control means refer to survey years before 1992 with wages in 2011 prices. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C4: Robustness of selection effects

	Log wages				
	Baseline	By birth year	By years of schooling	By survey year	Excluding sparse cells
	(1)	(2)	(3)	(4)	(5)
Log location share	−0.044***	−0.061***	−0.044***	−0.057***	−0.052***
	(0.002)	(0.003)	(0.002)	(0.003)	(0.008)
Observations	97,165	97,165	97,165	97,165	72,364
Control mean	9.20	9.20	9.20	9.20	9.12

Each column is a regression, and each observation is a male wage earner of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974. Wages are log hourly wages. Location shares are calculated within groups defined by birth district, cohort, and schooling, pooling years of schooling into three bands and pooling across survey years. Column (1) reports the fixed-effects estimate of table 5. Column (2) computes location shares by birth year rather than cohort. Column (3) computes location shares using exact years of schooling rather than pooling into bands. Column (4) computes location shares by survey year rather than pooling across years. Column (5) excludes sparse cells with fewer than five observations. I include fixed effects for birth district by cohort by years of schooling, as well as fixed effects for cohort by destination by survey year. I cluster standard errors by birth district. Control means refer to unexposed cohorts. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Counterfactuals

I describe how I solve the model and report additional results.

D.1 Solving the model

I derive exact-hat expressions with notation $\hat{x} = x'/x$ in changes for counterfactual x' and baseline x. As in Dekle et al. (2008), I express counterfactuals as changes relative to the observed equilibrium. The goal is to characterize how production responds to changes in school construction, accounting for equilibrium changes in wage rates. I solve the market-clearing conditions separately for each survey year, drawing on the sample of male wage earners of age 2 to 6 or 12 to 17 in 1974. I proceed in six steps, suppressing survey year t.

First, I recover the baseline quantities. Wage rates $r_{k\ell}$ are recovered from the cohort-destination-year fixed effects of equation 28, as derived in appendix C. These wage rates are estimated conditional on θ. For each birth district and cohort, I recover location choice probabilities as a function of schooling. I match each group's observed location shares, pooling across survey years while allowing the probability of leaving home to vary with schooling according to λ. Conditional on leaving, destination probabilities are given by observed destination shares, pooled across schooling and survey years. Unobserved routes remain unavailable.

I recover human capital from observed wages. By equation 29, I adjust each individual's wage for the wage rate in their destination and for selection, calculating $\exp(\log w_i - \log r_{k\ell})\pi_{i\ell}^{1/\theta}$. I average these quantities within groups defined by birth district, cohort, and years of schooling, pooling across survey years. Human capital and location choice probabilities determine baseline labor supply and wage bills by equations 17 and 15. For each survey year, I recover production weights $\alpha_{k\ell}$ from cohort shares of the wage bill. Baseline schooling, location choice probabilities, and wage rates then anchor schooling costs, migration frictions, and productivity.

Second, school construction reduces schooling costs. Differencing equation 20,

$$\log c_i' - \log c_i = -\delta T_k \Delta S_j, \quad \Delta S_j = S_j' - S_j.$$

Changes in construction are given by the counterfactual of interest, exposure T_k is observed, and parameter δ is as estimated.

Third, to characterize responses, I consider candidate changes $\hat{r}_{k\ell}$ in wage rates and take them as given. Let $\Delta e_i = e_i' - e_i$ denote schooling changes and $\pi_{i\ell}$ denote baseline location choice probabilities. Under equation 4, the access term $(r_{k\ell}/\tau_{jk\ell})^\theta$ changes by $\hat{r}_{k\ell}^\theta \exp(\lambda \Delta e_i \mathbb{1}\{\ell \neq j\})$. It follows that

$$Z_i = \frac{\Omega_i'}{\Omega_i} = \sum_\ell \pi_{i\ell} \hat{r}_{k\ell}^\theta \exp(\lambda \Delta e_i \mathbb{1}\{\ell \neq j\}),$$

where baseline frictions cancel. Equations 6 and 8 give changes in migration and expected utility.

$$\pi_{i\ell}' = \frac{\pi_{i\ell} \hat{r}_{k\ell}^\theta \exp(\lambda \Delta e_i \mathbb{1}\{\ell \neq j\})}{Z_i}, \quad \frac{\bar{r}_i'}{\bar{r}_i} = Z_i^{1/\theta}$$

Schooling and migration respond jointly. Differencing the schooling first-order condition of equation 31, I obtain

$$\rho \Delta e_i = \delta T_k \Delta S_j + \frac{1}{\theta} \log Z_i + \log \frac{\rho + \lambda m_i'/\theta}{\rho + \lambda m_i/\theta},$$

where $m_i' = 1 - \pi_{ij}'$ and ability, skill, and baseline costs cancel. When the home probability is positive, the last two terms also equal $\log \hat{r}_{kj} + \log(\Lambda_i'/\Lambda_i)$, which correspond to the home and away terms of equation 10. I solve the schooling condition subject to $e_i' \geq 0$, allowing corner solutions. For individuals with zero baseline schooling, I use the cost that makes zero schooling satisfy the first-order equality.

Fourth, on the supply side, human capital changes by $\hat{h}_i = \exp(\rho \Delta e_i)$. Applying equation 17, cohort labor supply changes by

$$\hat{H}_{k\ell}^S = \sum_{i \in \mathcal{I}_k} \omega_{i\ell} \exp(\rho \Delta e_i) \left(\frac{\pi_{i\ell}'}{\pi_{i\ell}} \right)^{1-1/\theta}, \quad \omega_{i\ell} = \frac{h_i \pi_{i\ell}^{1-1/\theta}}{H_{k\ell}},$$

where the sum is over observed routes. The weights are baseline shares of cohort labor supply in each destination. They also equal shares of the cohort's destination wage bill because each unit of human capital earns the same wage rate $r_{k\ell}$.

Fifth, on the demand side, productivity and production weights are fixed. Equations 13 and 14 imply

$$\hat{H}_\ell = \prod_k \hat{H}_{k\ell}^{\alpha_{k\ell}}, \quad \hat{r}_{k\ell} = \frac{\hat{H}_\ell^\kappa}{\hat{H}_{k\ell}}. \tag{32}$$

Wage rates depend on both aggregate human capital and cohort composition.

Sixth, I evaluate counterfactual changes in construction. I consider candidate changes in wage rates, and I compute schooling, migration, and labor supply as above. I adjust candidate wage rates until the labor supply changes satisfy equation 32 in every cohort and destination. These conditions in changes are equivalent to market clearing in levels because markets also clear in the baseline equilibrium. I select equilibria connected to the observed equilibrium by gradually shifting construction from observed to counterfactual levels. Output changes then follow from

$$\hat{Y}_\ell = \hat{H}_\ell^\kappa = \hat{r}_{k\ell}\hat{H}_{k\ell}.$$

By equation 15, aggregate output is given by the aggregate wage bill. I allocate this wage bill by birthplace for people and by work location for places, weighting survey years equally. For inequality, let W_j denote the district wage bill and N_j the baseline district population. District mean wages are $\bar{w}_j = W_j/N_j$, and the population-weighted Gini coefficient is

$$G = \frac{\sum_j \sum_m N_j N_m |\bar{w}_j - \bar{w}_m|}{2\left(\sum_j N_j\right)\left(\sum_j W_j\right)}.$$

I calculate counterfactual Gini coefficients using counterfactual wage bills, holding baseline populations fixed as both denominators and weights.

Table D1: Migration responses

		Data	
	Model	Estimate	SE
Total migration	1.23	1.23	(0.54)
From sparse origins	0.45	0.35	(0.34)
From dense origins	0.78	0.88	(0.43)
To sparse destinations	0.80	0.67	(0.53)
To dense destinations	0.43	0.56	(0.33)

Each row is an outcome, and each observation is a male wage earner of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974. Entries report effects of an additional school per 1,000 children on migration in percentage points. Migration is residence outside of the birth district. Origin and destination rows interact migration with indicators for being above or below the median in the distribution of 1971 population density across districts. Each pair of outcomes sums to total migration. Total migration is targeted in estimation, while its sparse and dense components are untargeted.

Table D2: Parameter sensitivity

(a) Equilibrium

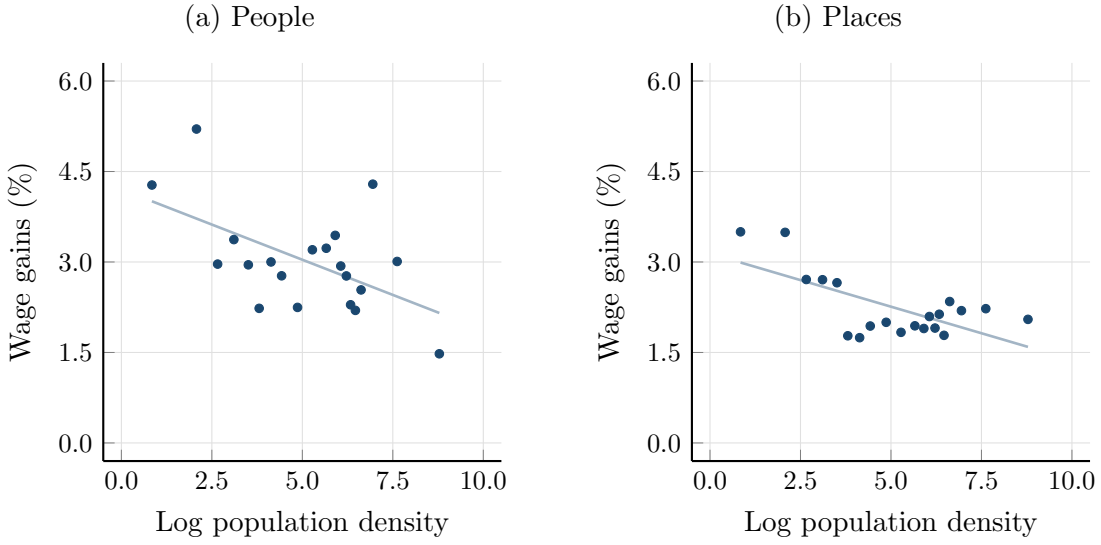
	Output	Inequality (%)	
κ	(%)	People	Places
0.85	2.58	−1.20	1.26
0.6	1.44	−1.12	0.39
0.7	1.84	−1.14	0.59
0.8	2.32	−1.17	0.96
0.9	2.93	−1.23	1.78
1.0	3.91	−1.38	4.70

(b) Selection

	Output	Inequality (%)	
θ	(%)	People	Places
22.9	2.58	−1.20	1.26
10	2.66	−1.23	1.49
15	2.62	−1.23	1.39
20	2.59	−1.21	1.30
25	2.57	−1.19	1.23
30	2.56	−1.17	1.18

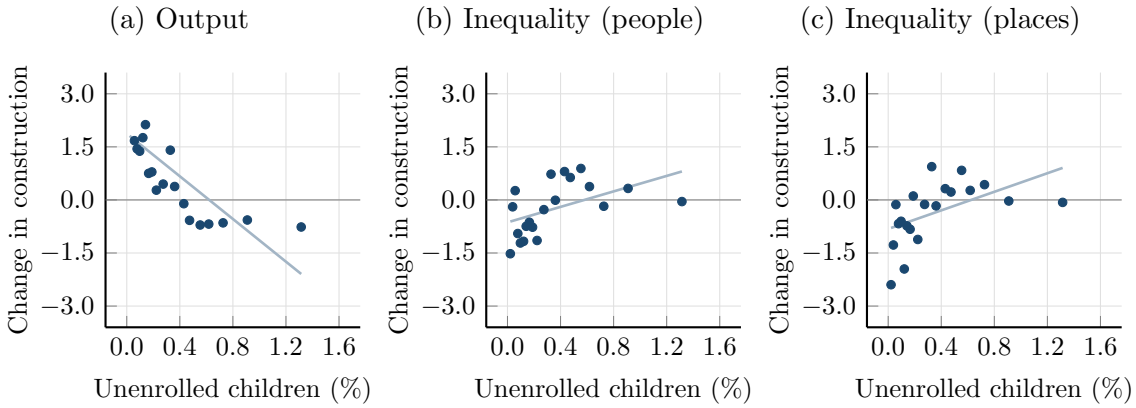
Each row reports the impacts of observed school construction relative to a baseline of no school construction. The top rows show the estimated values of (κ, θ). Panel (a) varies κ, and panel (b) varies θ. In each case, I refit (δ, ρ, λ) to the schooling, wage, and migration effects of table 3. I report impacts on output and inequality in percentage changes. Output is aggregate wages. Inequality is the population-weighted Gini coefficient of district mean wages. Inequality across people compares mean wages of people born in different districts. Inequality across places compares mean wages of people living in different districts. The sample is male wage earners of age 2 to 6 (exposed) or 12 to 17 (unexposed) in 1974.

Figure D1: Historical incidence



Each panel is a binned scatter plot across the 282 program districts. The y-axis is the percentage wage gain from observed school construction, relative to no construction. Gains for people are those of people born in a district. Gains for places are those of people living in a district. The x-axis is log population density in 1971.

Figure D2: School allocations



Each panel is a binned scatter plot across the 282 program districts. Panels correspond to the three objectives in figure 3, allowing 20% of the national budget to move across districts. Output is aggregate wages. Inequality across people compares mean wages of people born in different districts. Inequality across places compares mean wages of people living in different districts. The y-axis is the change in school construction per 1,000 children, relative to observed construction. The x-axis is a district's share of all unenrolled children of primary school age in 1971.